

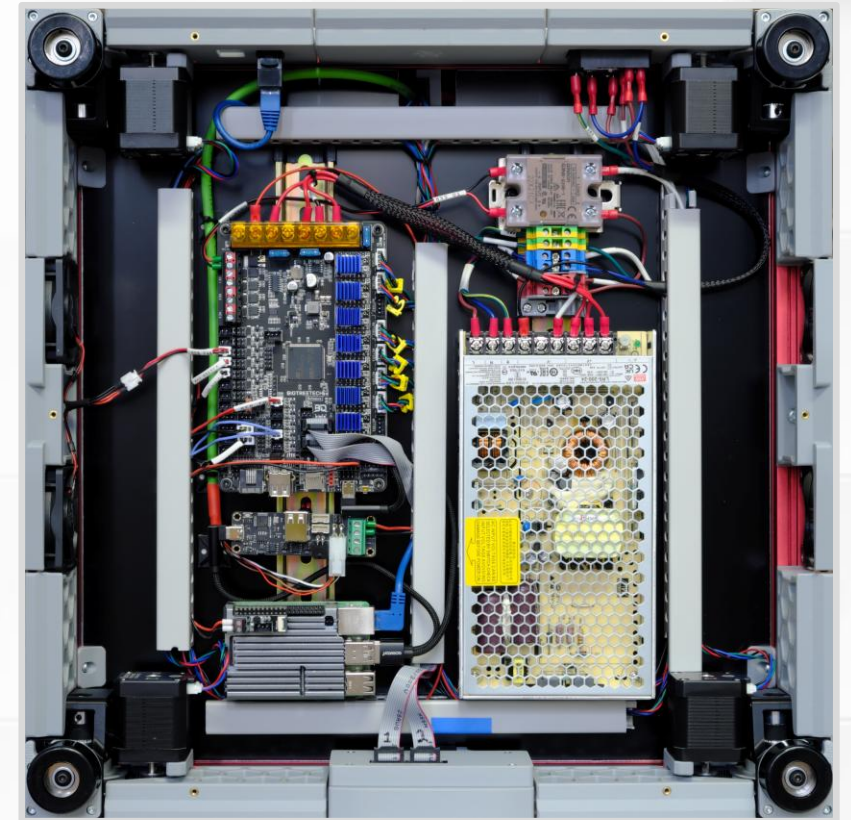
An Introduction to 3D Printing

Part 1: FFF Printers

Introduction, Process, Limitations, and Advantages

About Me

- Woodworking and furniture making.
- Built my own printer to learn about 3D printing.
- 6k hours (250 days) of print time.
- [Printables.com](https://www.printables.com)
Flying Gyroscope



(Voron 2.4 GitHub)

FAQ: Which Printer?

Which printer should I get?

- Is the printer the hobby, or is making things the hobby?
- Get the most **reliable** printer you can afford.
 - Do not get sucked into marketing.
 - Do not obsess over spec sheets or speeds.
- Top brands give you excellent slicer profiles out of the box.
 - Very highly recommended for beginners.

FAQ: Tuning

How do I tune my printer?

- Keep it simple to help diagnose problems.
 - Practice restraint when changing slicer settings.
 - Keep the printer stock, unmodified.
- Make one change and try A vs B test.
 - Can you actually see or measure a difference?
- [Ellis' Print Tuning Guide](#).
- [OrcaSlicer Calibration Walkthrough](#).

Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

Thermal Expansion

3D Printing is a Process

1) Start with a 3D design (digital file).

- If you download, check the license for restrictions.
- **STL**. Approximates with tiny triangles.
- **STEP**. Real geometry. Great for 3D modeling.
- **3MF**. Container, like a zip file.

Might contain multiple objects, slicer settings, multi-color settings, ...

2) Slice the design.

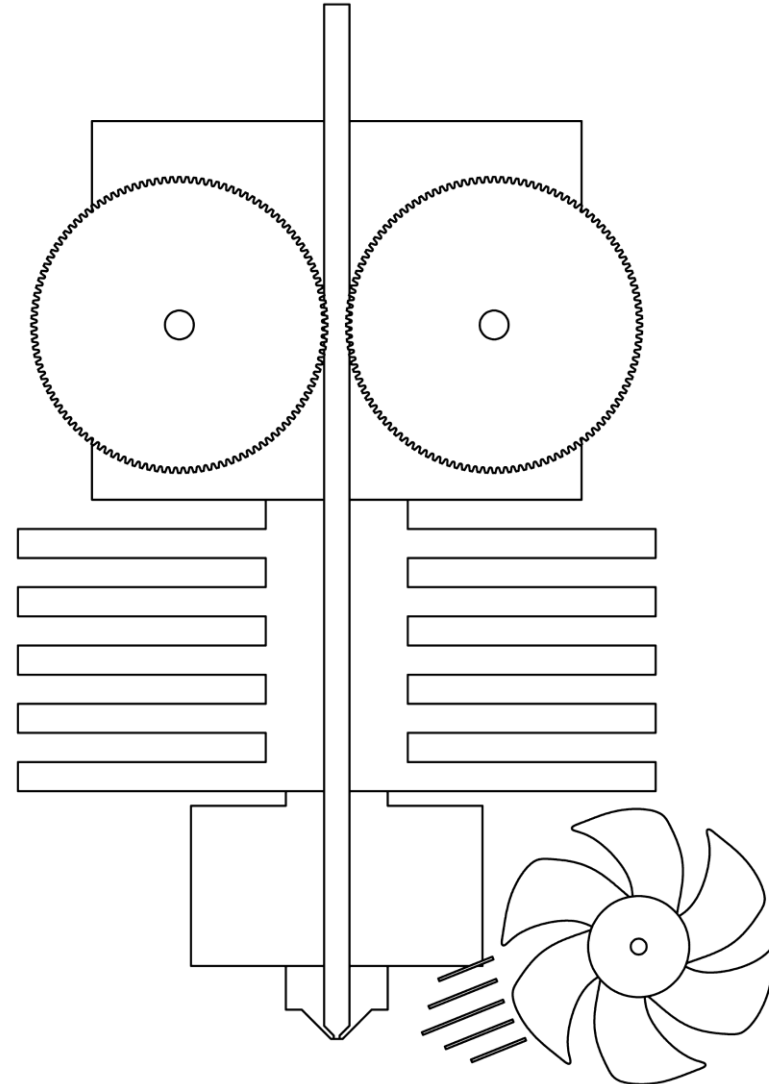
- Printers do not accept 3D models.
- Use software to “slice” objects into vertically-stacked layers.
- Translate shapes into Gcode. Standardized commands.

3D Printing is a Process

- 3) Print. Do you need to ...
 - Modify the design?
 - Tune printer settings?
- 4) Post-process.
 - Remove supports.
 - Assemble.
 - Glue.
 - Fill and sand to remove layer lines.
 - Paint.

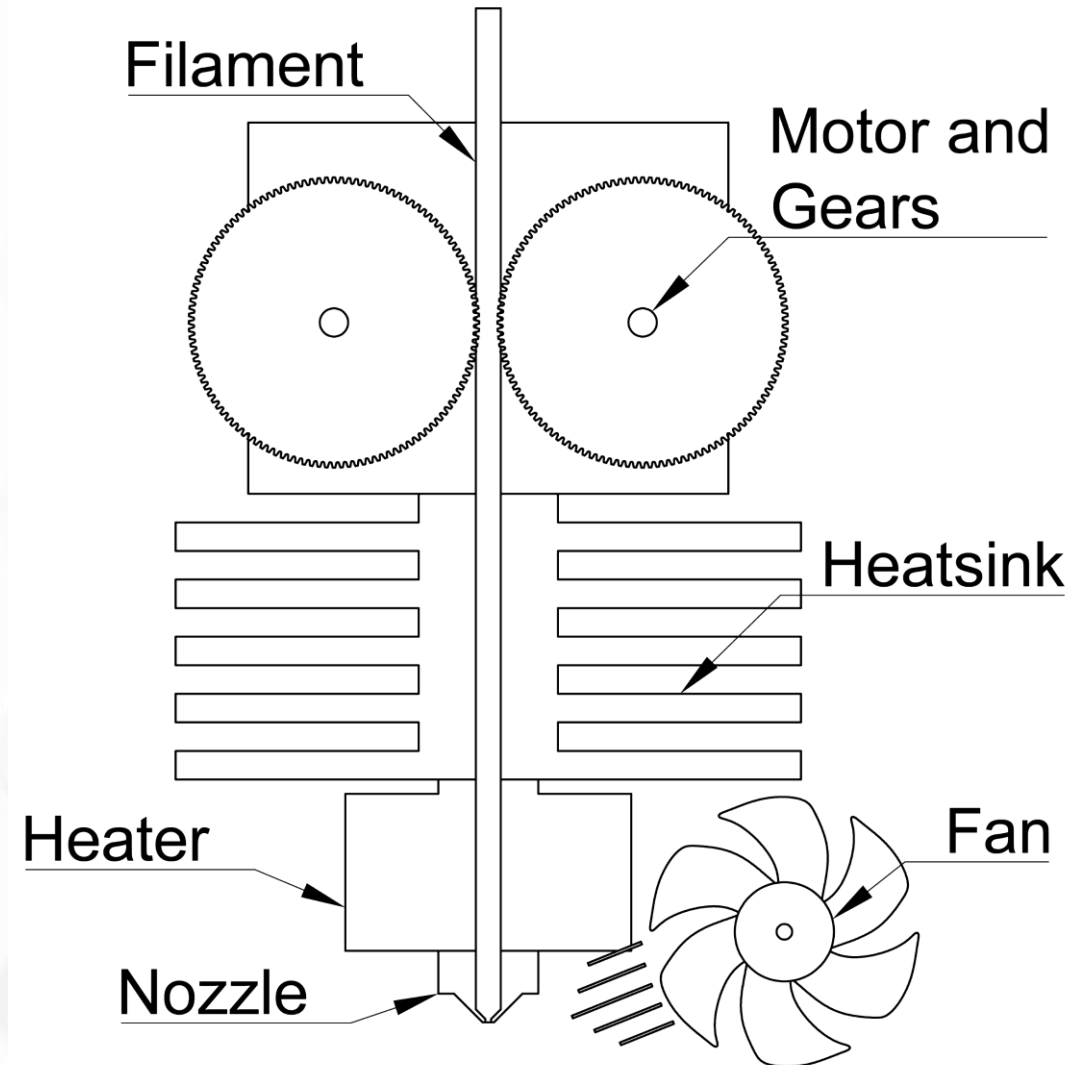
Pushing Plastic

- What kind of 3D printing?
- **FFF. Fused filament fabrication.**
- FDM™. Fused deposition modeling.
- Filament and extrusion-based additive manufacturing.
- Toolhead. Moves around and deposits plastic.



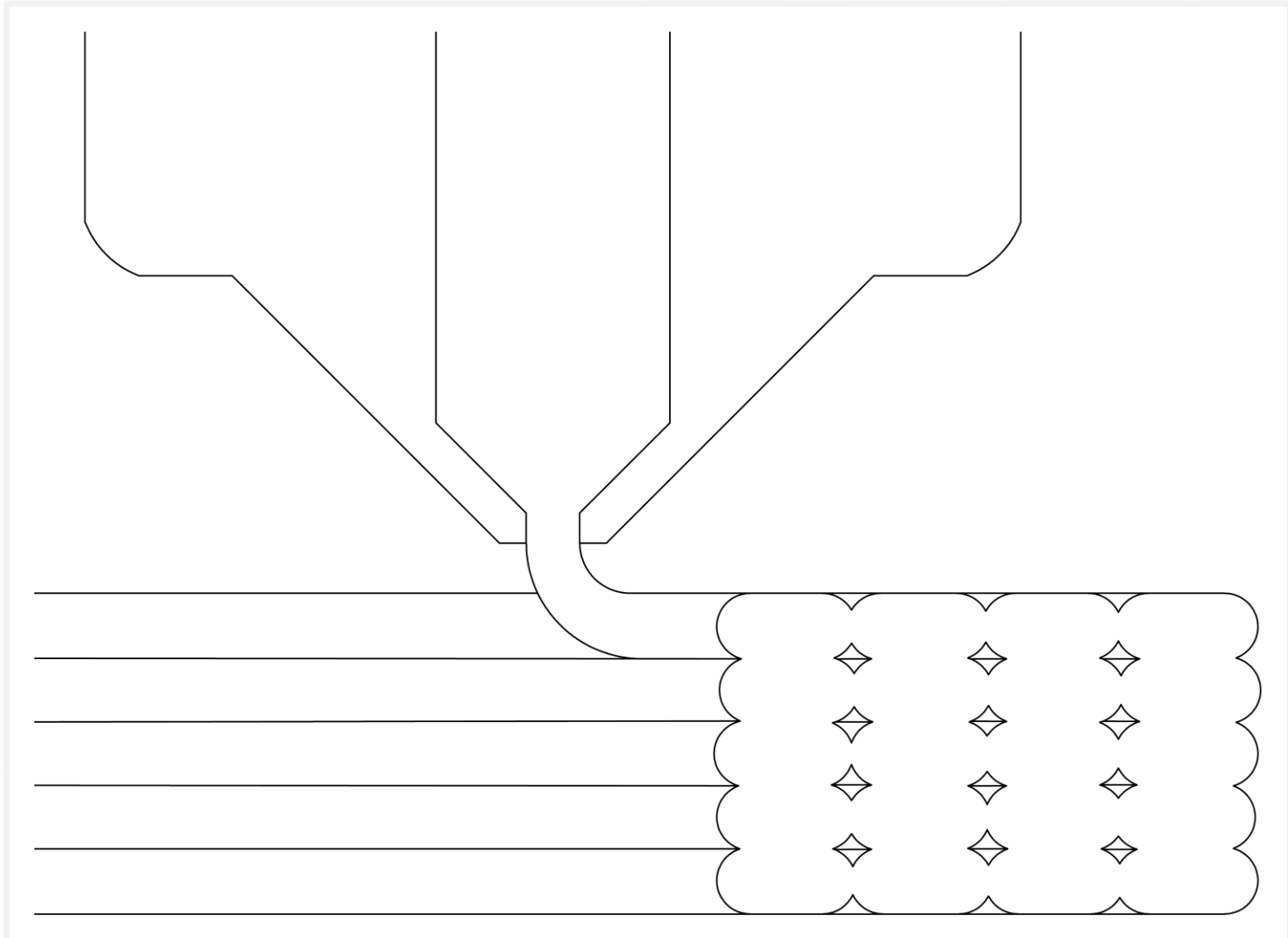
Pushing Plastic

- Filament. Long, continuous plastic noodle.
- Motor and gears push filament.
- Heatsink keeps heat in the heater and nozzle.
- Heater softens filament.
- Nozzle opening determines line width.
- Fan cools hot plastic.



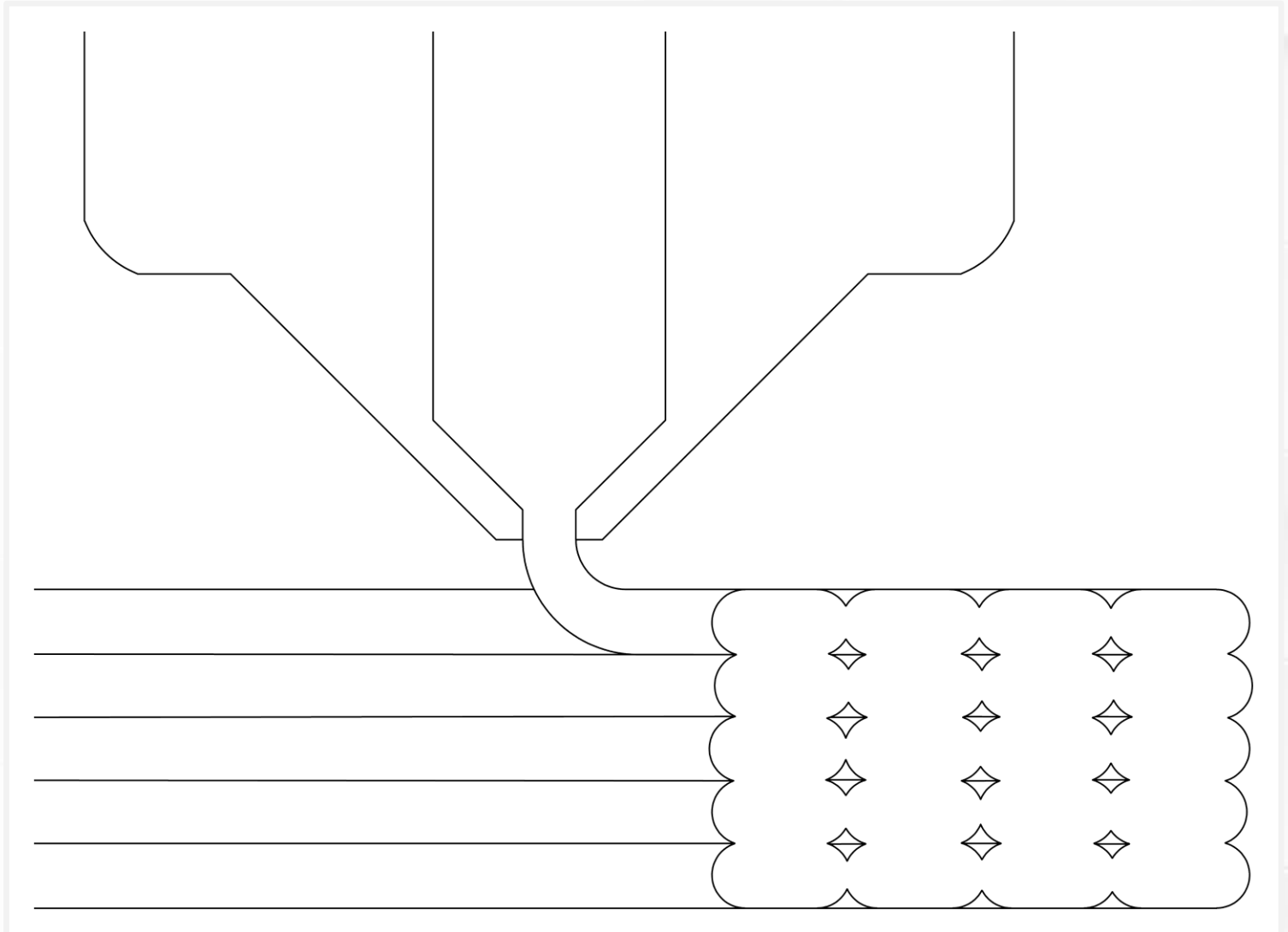
Extrusions

- Toolhead deposits a line called an **extrusion**.
- Best to squish extrusions into place.
- Extrusions lock in place when they cool.
- Printing needs fast cold-hot-cold thermal cycle.



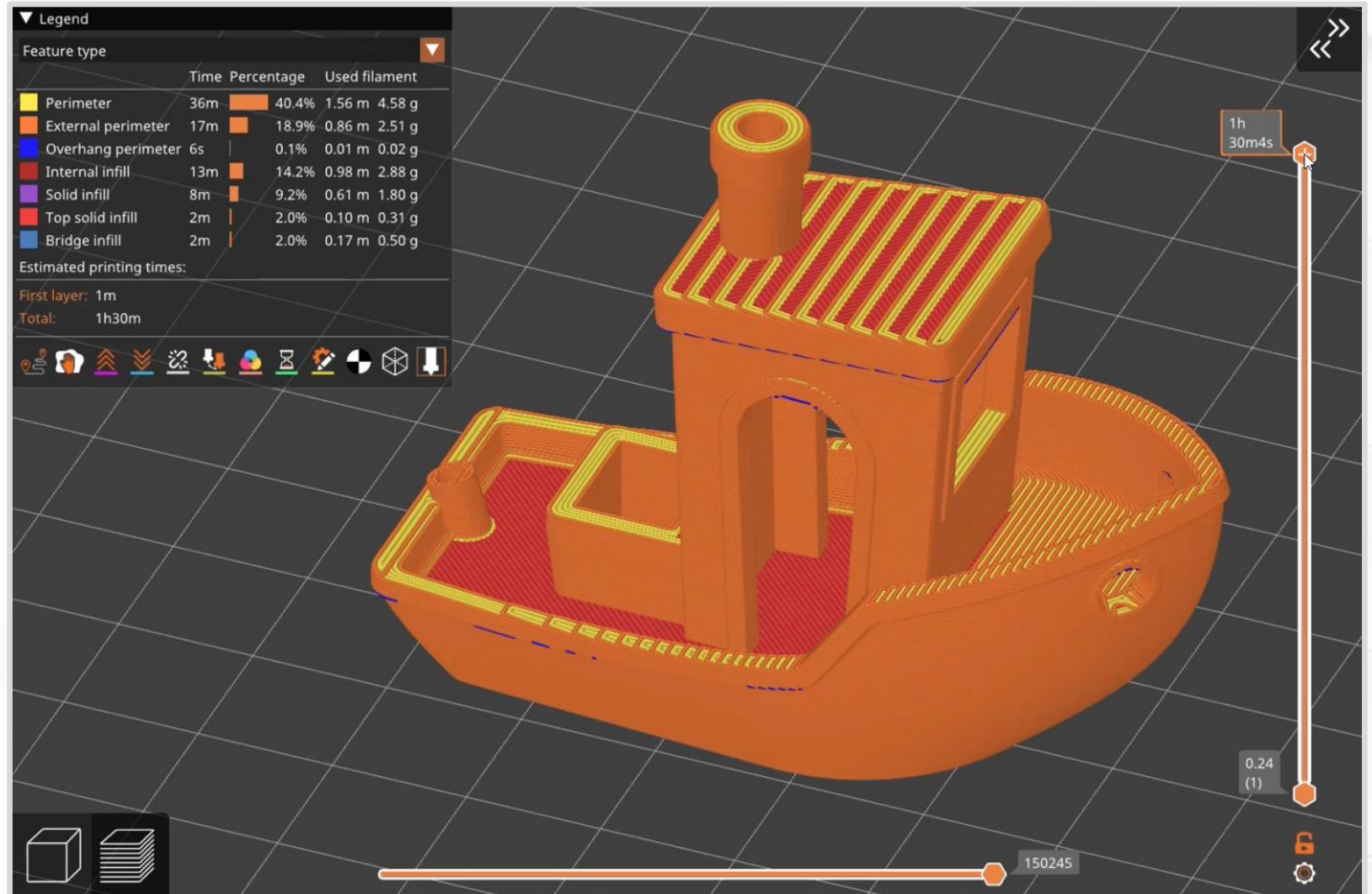
Stacking Layers

- Printers *simulate* 3D shapes by stacking layers.
- You could call it 2.5D printing.



Slicer Layers

Let's see how
PrusaSlicer handles
layers.

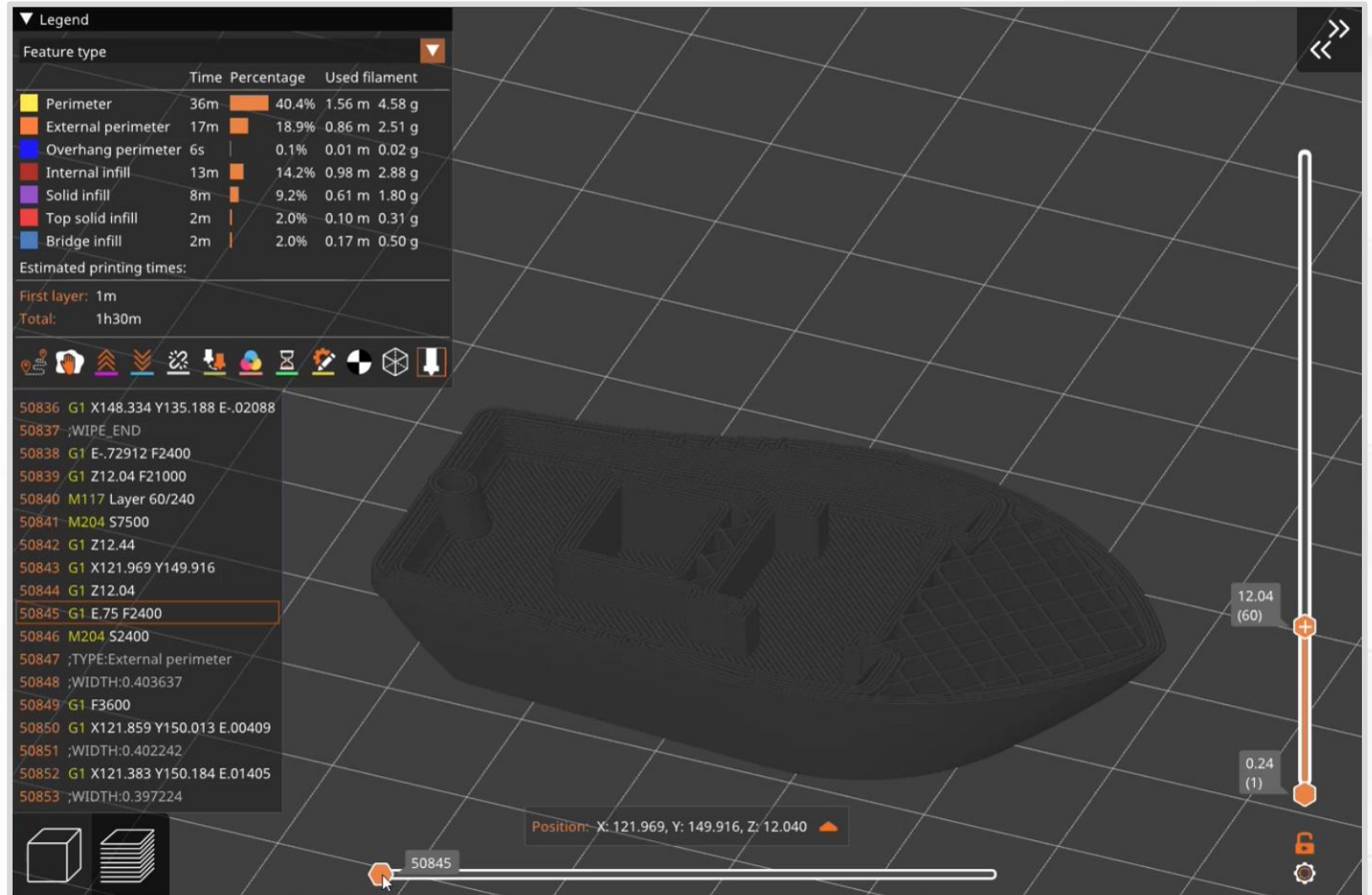


(Animation,
FlyingGyroscope GitHub)

Slicer Layers

- Single layer preview.
- Gcode shows what the printer is actually doing (lower left).
- *G1* = move. Can include X, Y, or Z coordinates.
- *G1 X Y E* = move and extrude.

(Animation,
[FlyingGyroscope GitHub](#))

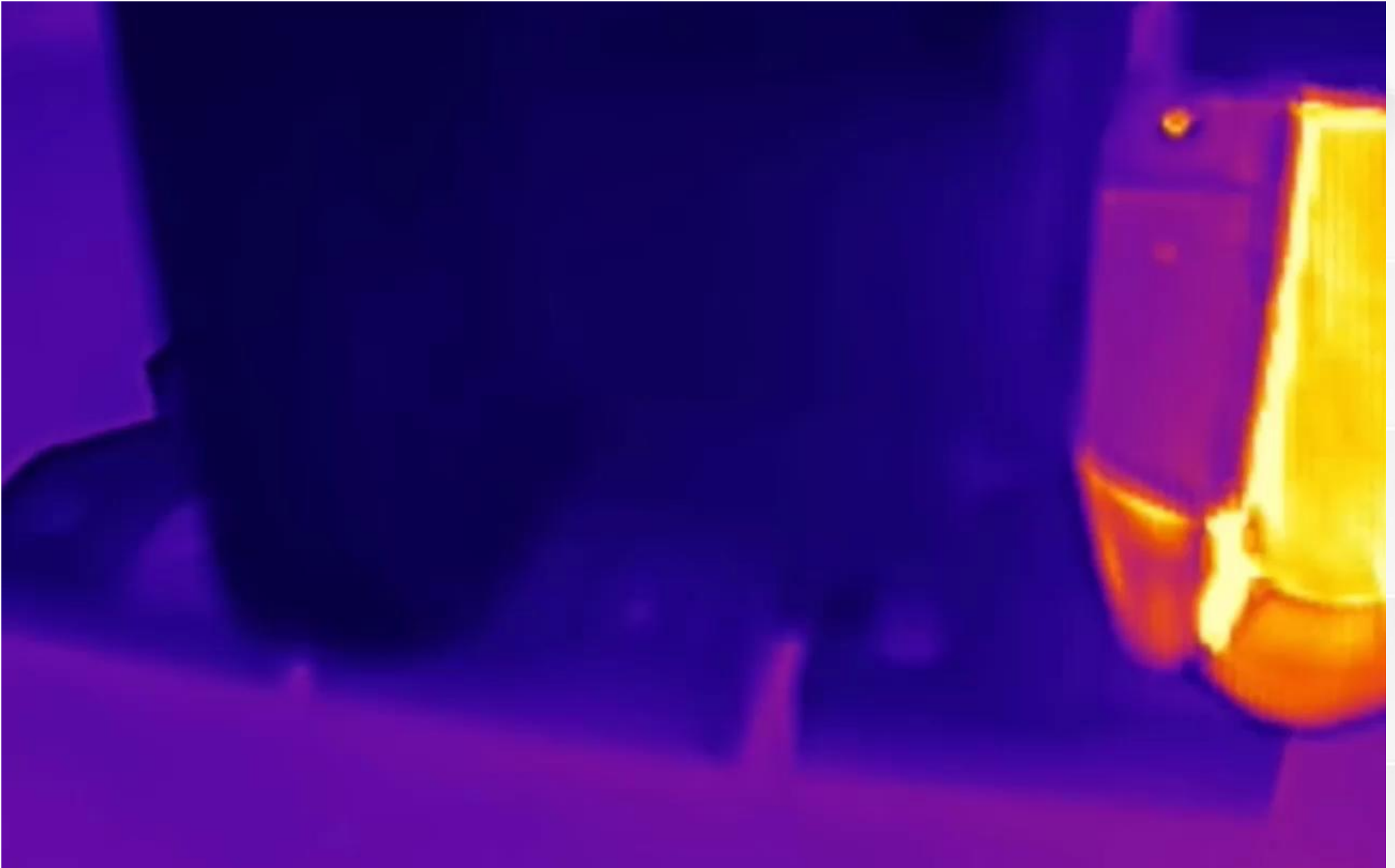


Slicer Layers

- Slicers sort and categorize extrusions.
- Adjust settings for each category. Typically ...
 - Printing speed.
 - Cooling (fan speed).

▼ Legend					
Feature type ▼					
		Time	Percentage	Used filament	
	Perimeter	36m		40.4%	1.56 m 4.58 g
	External perimeter	17m		18.9%	0.86 m 2.51 g
	Overhang perimeter	6s		0.1%	0.01 m 0.02 g
	Internal infill	13m		14.2%	0.98 m 2.88 g
	Solid infill	8m		9.2%	0.61 m 1.80 g
	Top solid infill	2m		2.0%	0.10 m 0.31 g
	Bridge infill	2m		2.0%	0.17 m 0.50 g
Estimated printing times:					
First layer: 1m					
Total: 1h30m					

(Animation,
Thermal camera
video by
NeedItMakelt)



Thermal Cycle

- Melting filament is technically incorrect. Why?
 - No transition into a liquid.
- Filament is **softened** into a viscous state.
 - Like a gel.
- Glass transition temperature instead of melting temperature.

Easy Filaments

- **PLA** (derived from corn starch).
 - General-purpose filament for indoor applications.
 - Excellent bridging.
 - Very stiff.
 - Deforms permanently under load (creep).
 - Low temperature resistance — will not survive inside a car on a hot summer day.
- **PETG** (plastic water bottles).
 - Pretty good general-purpose filament.
 - Hygroscopic (needs low-humidity storage).
 - Excellent choice for springs (elastic).
 - Moderate temperature resistance — can be used outdoors.

Easy Filaments

- What makes PLA and PETG easy?
 - Low warp (low shrink).
 - High dimensional accuracy.
 - Room temperature print chamber.

Proper Protocol

Let's review some everyday guidelines.

- Secure the loose end of a spool of filament **at all times**.
 - Tangles ruin prints and spools, and waste time.
- Do not reach into the printer while it is running.
 - Use the touchscreen or buttons to pause and resume prints.
- Watch the first layer before you walk away.
- Avoid burns.
 - Read nozzle and print bed temperatures on the display.
 - Never reach near a hot nozzle (210 °C = 410 °F).
- Do not leave fingerprints on print sheets.

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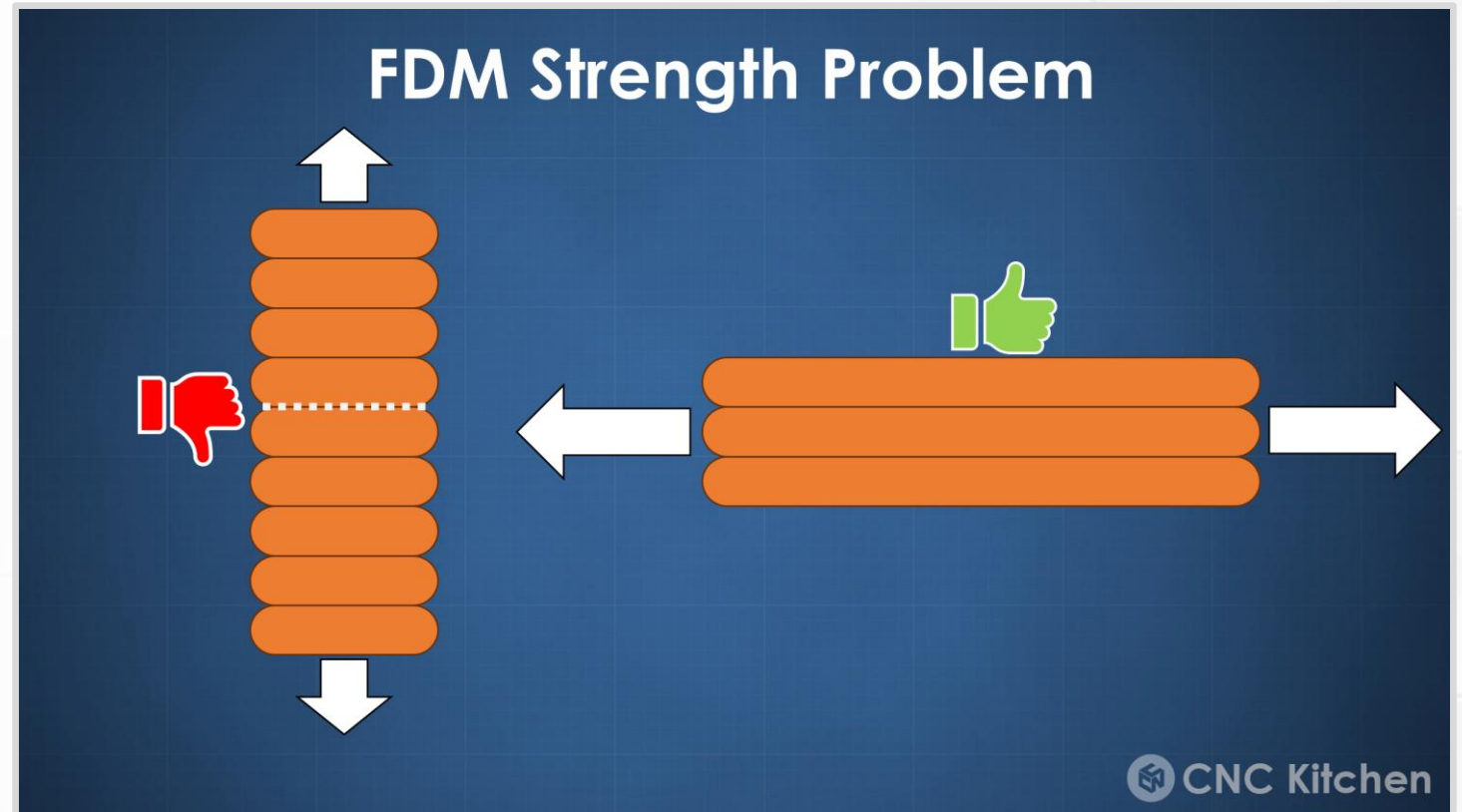
Thermal Expansion

Thermal Cycle

- Some thermoplastics shrink as they cool.
 - Up to 1 or 2%.
 - Seems small, but it can distort and skew dimensions.
 - Cracking and delamination in extreme cases.
- Advice?
 - Tightly control ambient temperature and cooling rate.
 - Some filaments (ABS, Nylon) like it hot → extra heating.
 - Some filaments (PLA) like it cold → extra cooling.
 - PETG is somewhere in-between.

Anisotropic Strength

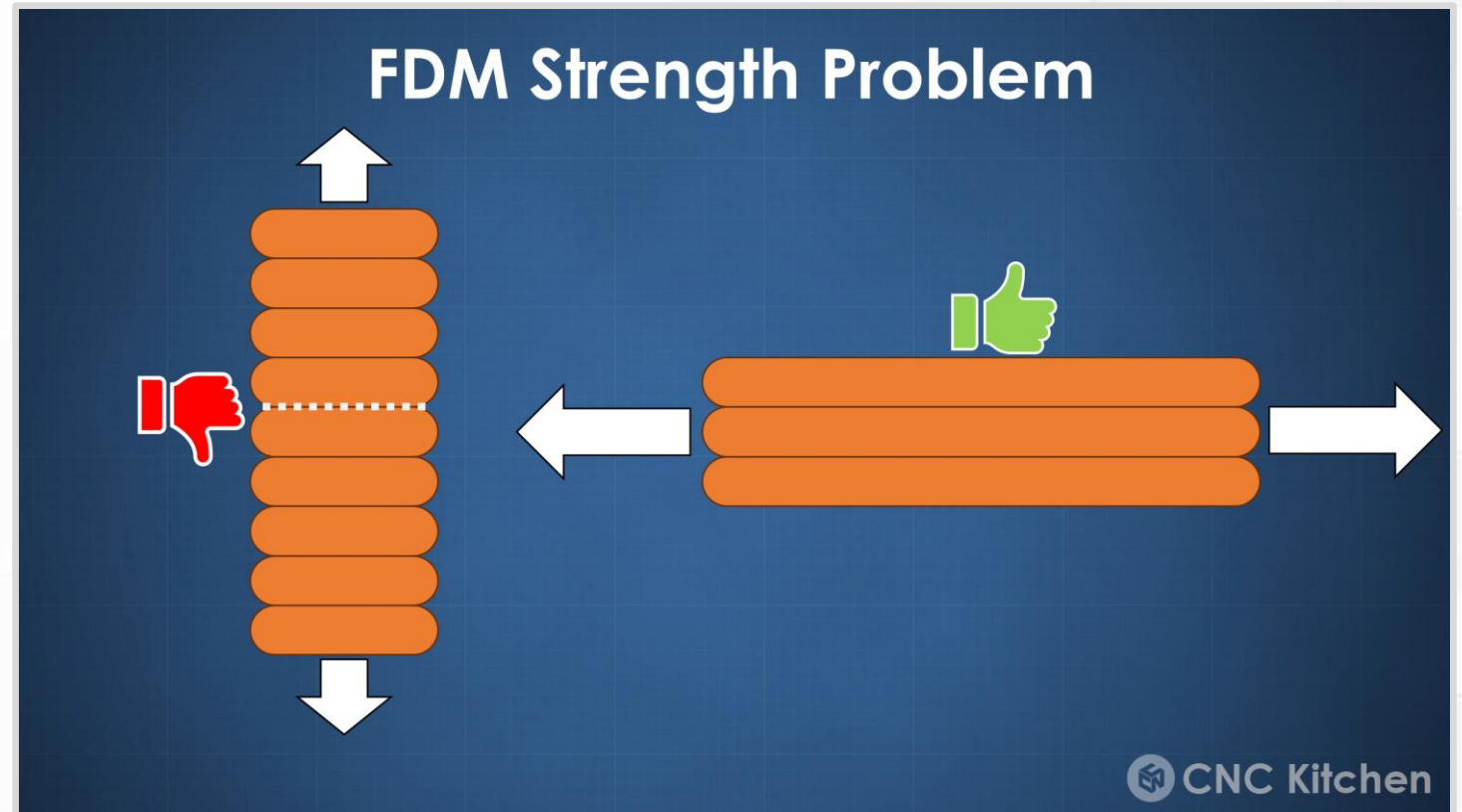
- Print orientation and layer lines affect strength.
- Layer-to-layer bonding is a weak point!
- Within an extrusion, hot filament bonds to hot filament.
- For a new layer, hot filament bonds to a warm layer.



(Picture from [CNC Kitchen](#))

Anisotropic Strength

- The “wrong” orientation can lose 50% of strength.
- Advice?
- Avoid tall and skinny shapes.
- Improve layer bonding.
 - Increase printing temps.
 - Decrease printing speed.



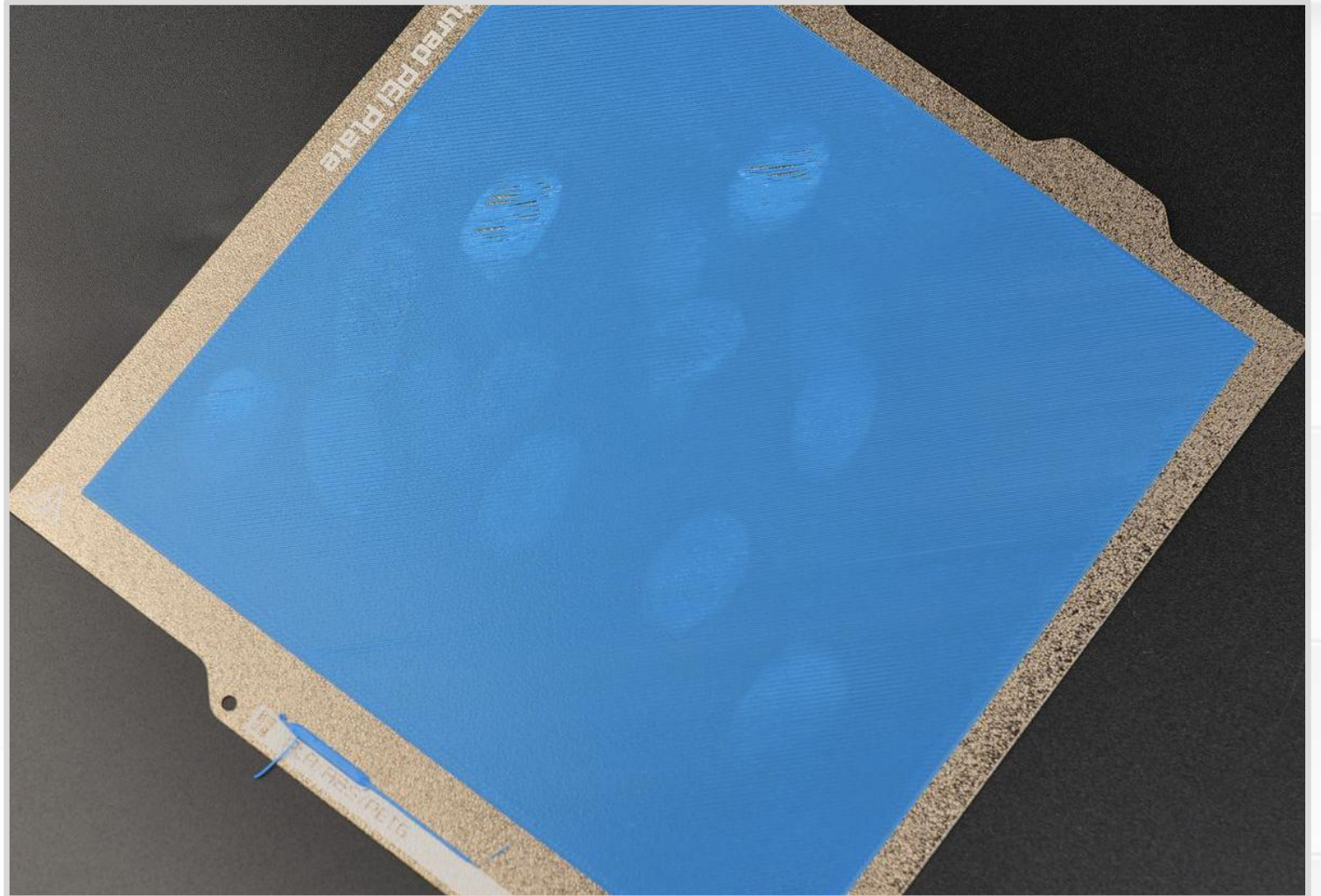
(Picture from [CNC Kitchen](#))

First Layer

- What is the most important layer in a 3D print?
 - The first layer!
 - First layers keep everything locked in place.
- Advice?
 - **Always watch the first layer.**
 - Tune first layer squish.
 - Flat surface area on the bottom of the 3D object.
 - Heated print bed.
 - Perfectly clean print bed.
 - Apply adhesion promoters (glue).

First Layer Mistakes

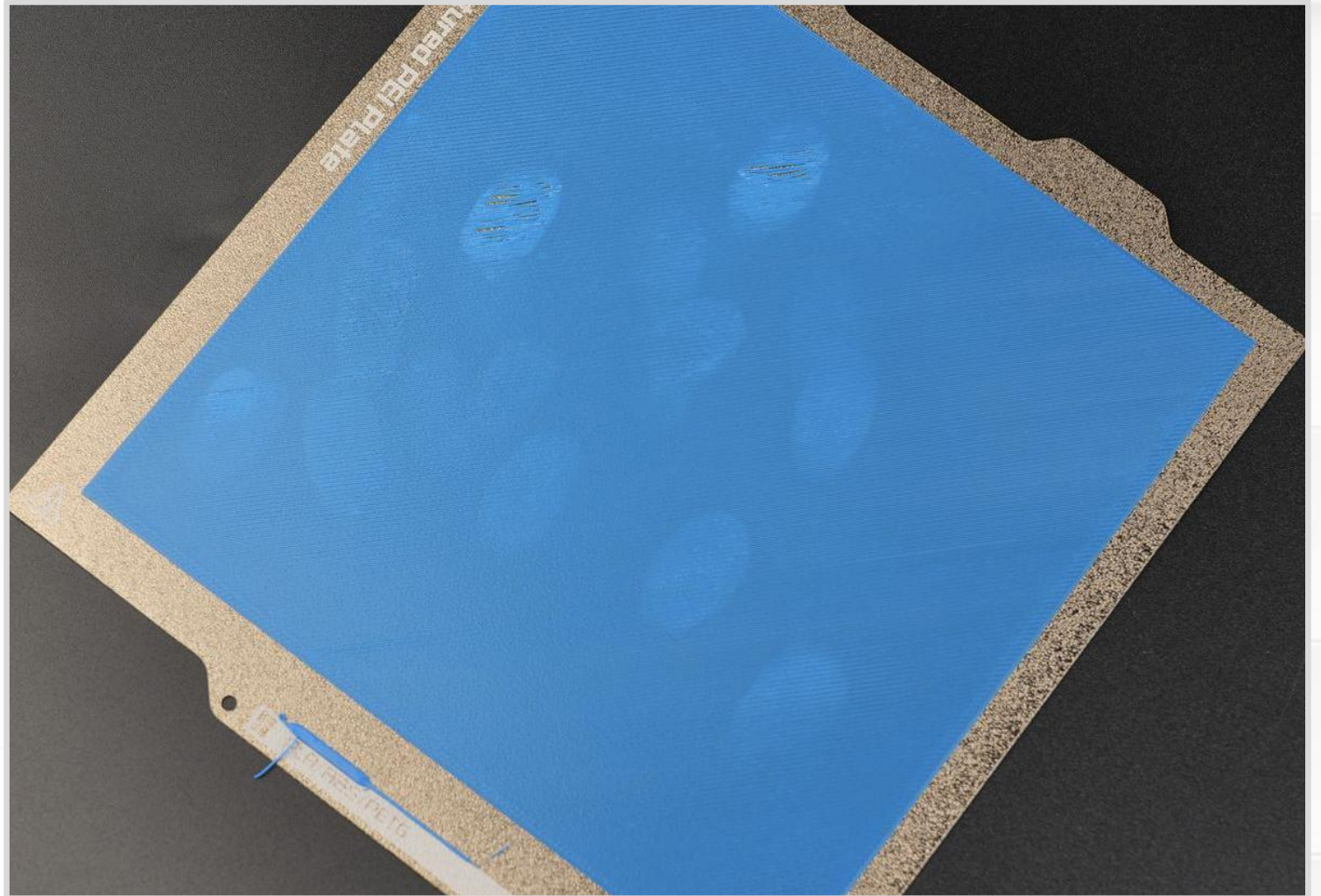
- What is wrong with this picture?
- **Fingerprints!** Plastic does not stick to oily fingerprints.



(Picture from [Bambu Labs Wiki](#))

First Layer Mistakes

- Advice?
- Deep cleaning: dish soap, soft brush, and water.
- Everyday use: window cleaner with ammonia.



(Picture from [Bambu Labs Wiki](#))

First Layer Mistakes

- What happens when a first layer does not stick?
- The print breaks free and makes “spaghetti”.
- You can see when things went wrong.
- Some of the rectangular print finished.



(Screenshot from [Everything STEM](#))

Stair-Stepping

- Printing with layers → Vertical slopes become stair steps.
- Mostly cosmetic effect.
- Advice?
- Rotate objects to hide layer lines.
- Smaller layer height. Increases print time.



(Picture from [Hamburg University Research Project](#))

First Layer Warping

- Why do corners and edges peel up?
- Temperature gradient from top to bottom.
- Regions shrink at different rates → Internal stresses → Stresses concentrate in corners.
- Shrinkage is a percent → Large parts have the most trouble.



(Picture from [Crealitty Cloud Blog](#))

First Layer Warping

Consider this analogy.

- Imagine you want to remove a label or piece of tape.
- What do you do?
- Start peeling up the corner because that is the weak point.
- Same thing here.
- Physics found the weak point.



(Picture from [Creality Cloud Blog](#))

First Layer Warping

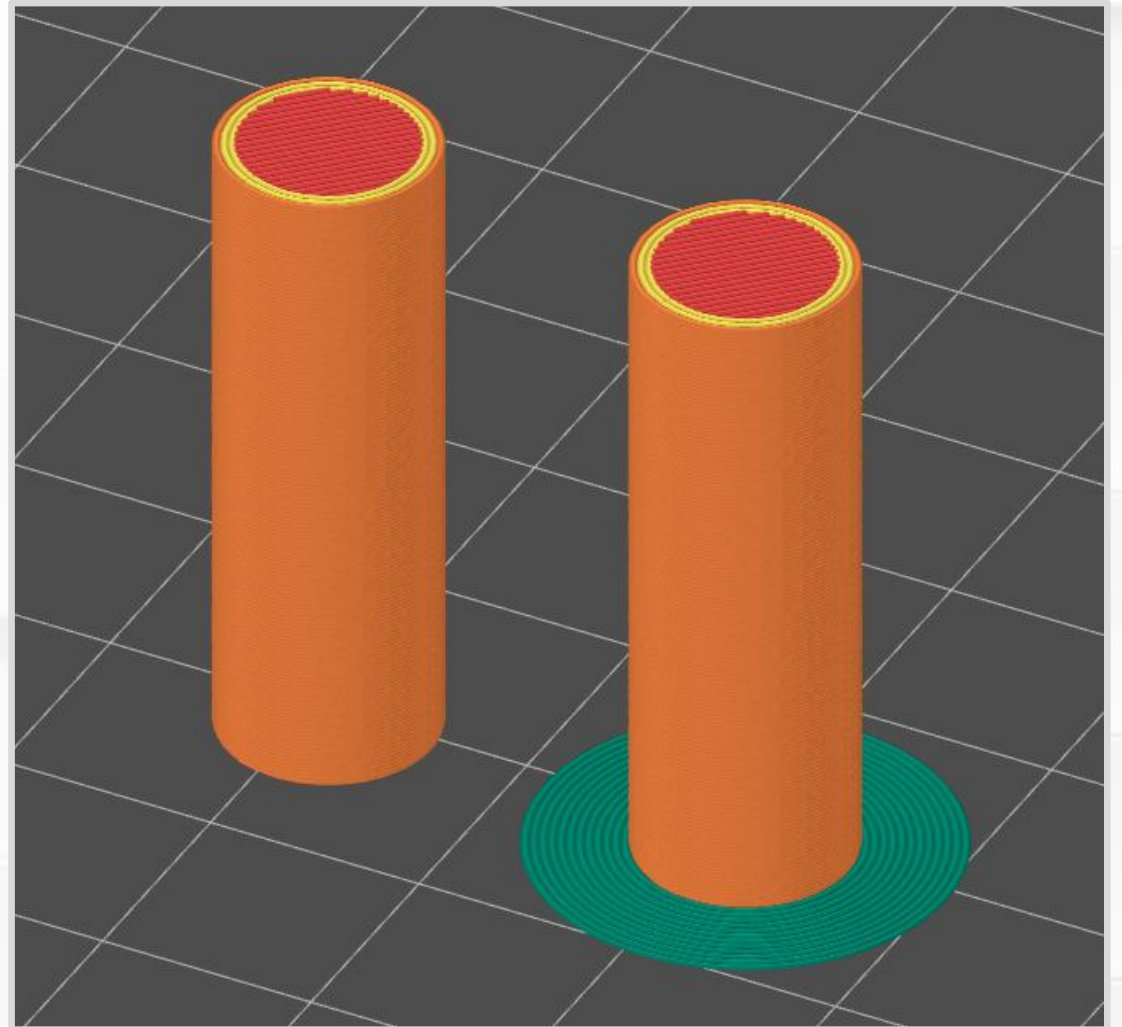
- Advice?
- Better control over temperatures.
 - Filament temperature.
 - Print chamber temperature.
 - Cooling fans.
- Evenly distributed cooling.
 - Do not want fans or drafts that cool one side.
- Round corners to reduce stress concentration.



(Picture from [Crealitty Cloud Blog](#))

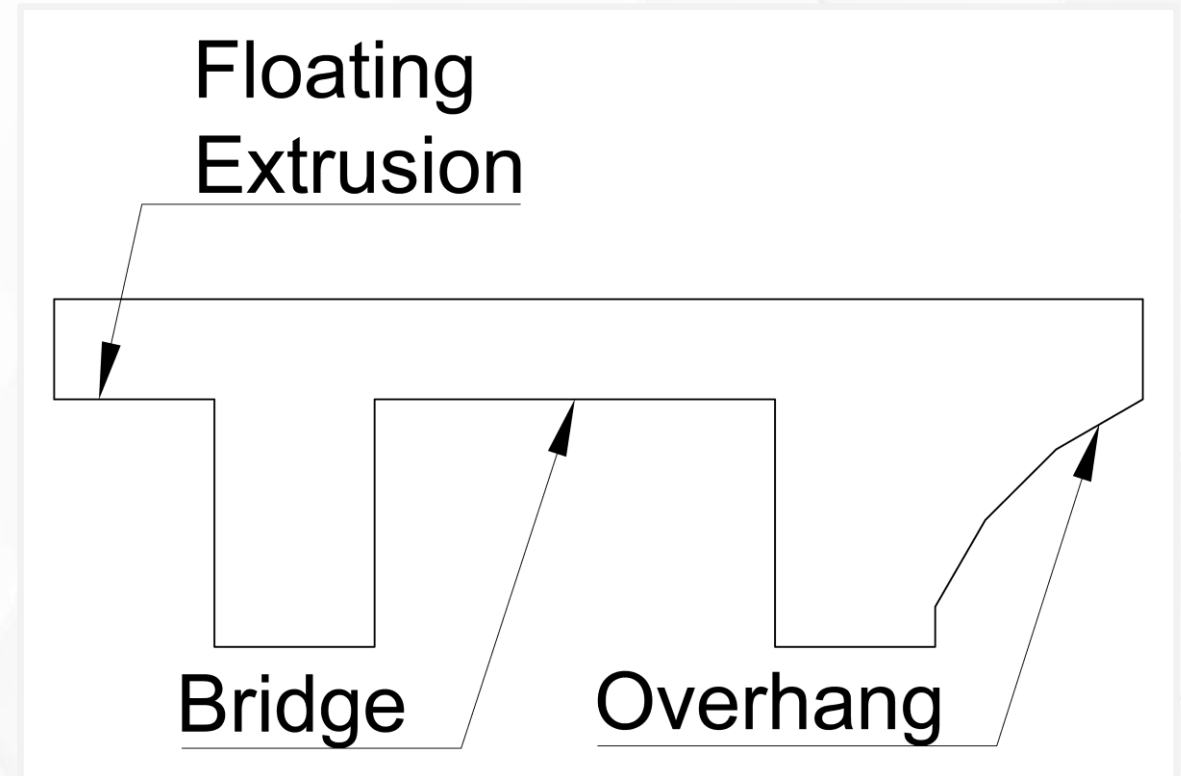
Brim

- Fights against first layer warping with extra bottom surface area.
- Concentric extrusions joined to bottom perimeter. (Green lines on the right.)
- Typically added with a slicer.
- One layer tall.
- Easily peeled away after printing.



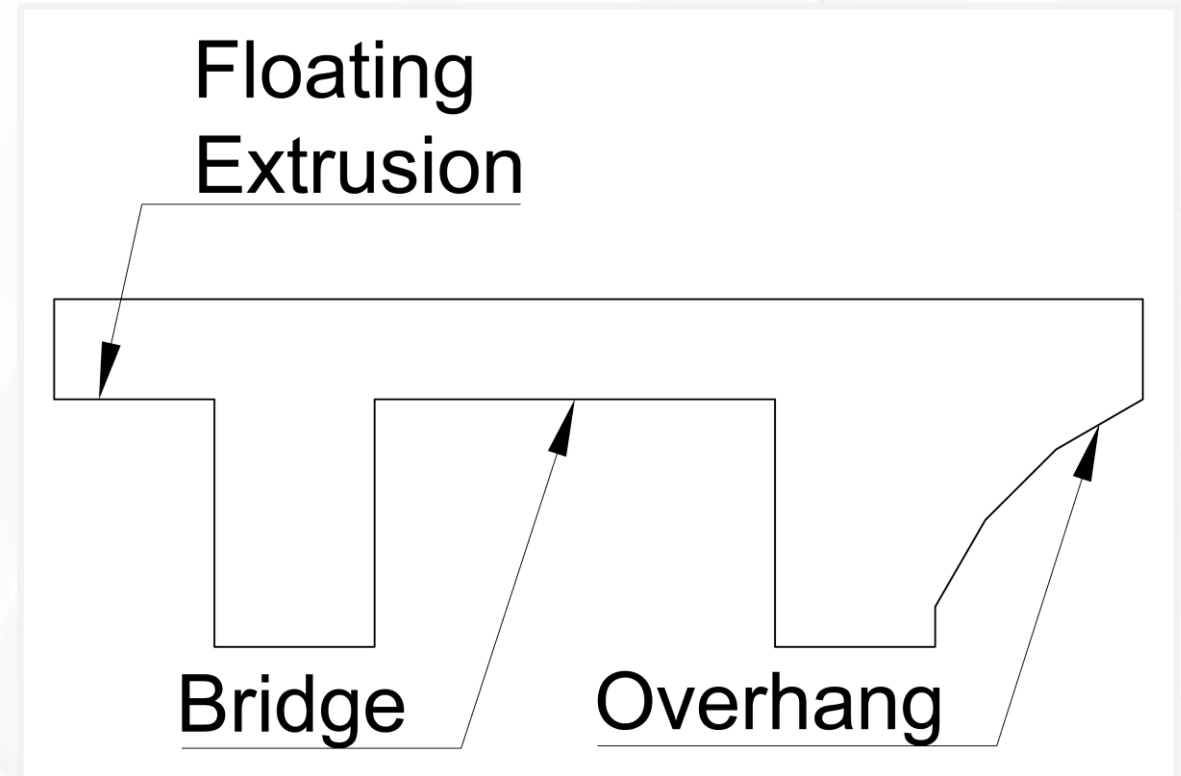
Levitating Layers

- *Critical thinking:* What happens when you try to levitate extrusions?
- Extrusions work best when you press them onto a solid surface.
- Predict what this print will look like.
- What parts will be the most deformed?



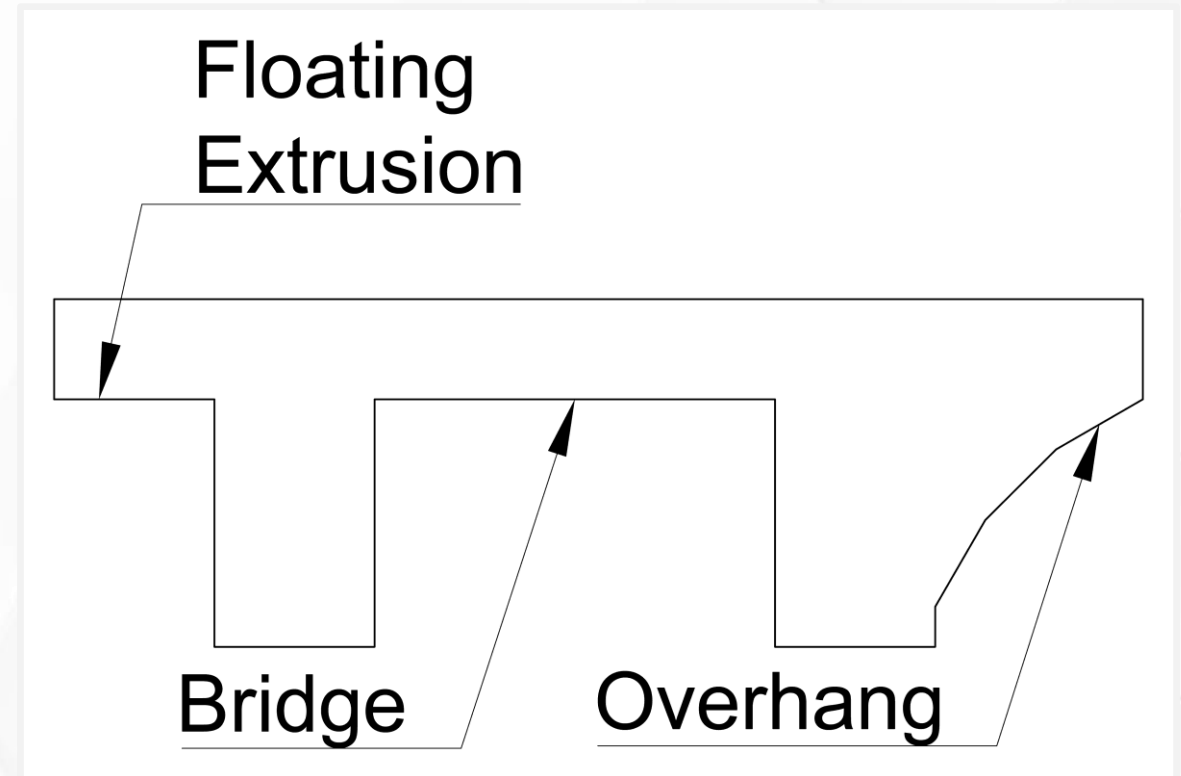
Overhang

- Extrusions that lean over an edge.
- Only the outermost extrusion is affected.
- Partially supported along one side.
- Advice?
- Keep overhangs at 45° to be safe.



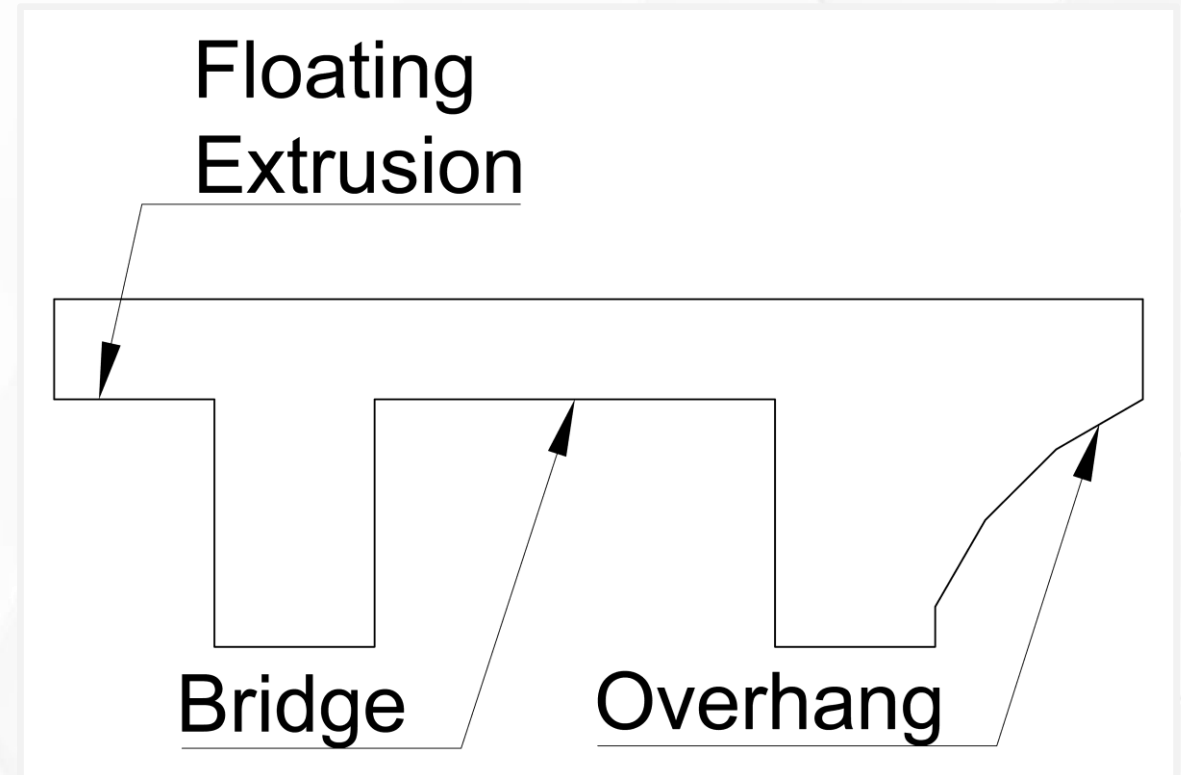
Bridge

- Both ends of an extrusion are supported and anchored.
- Middle hovers over a gap.
- The first few layers sag and droop down.
- Eventually recovers to flat layers.
- Advice?
- Keep bridges as short as possible.



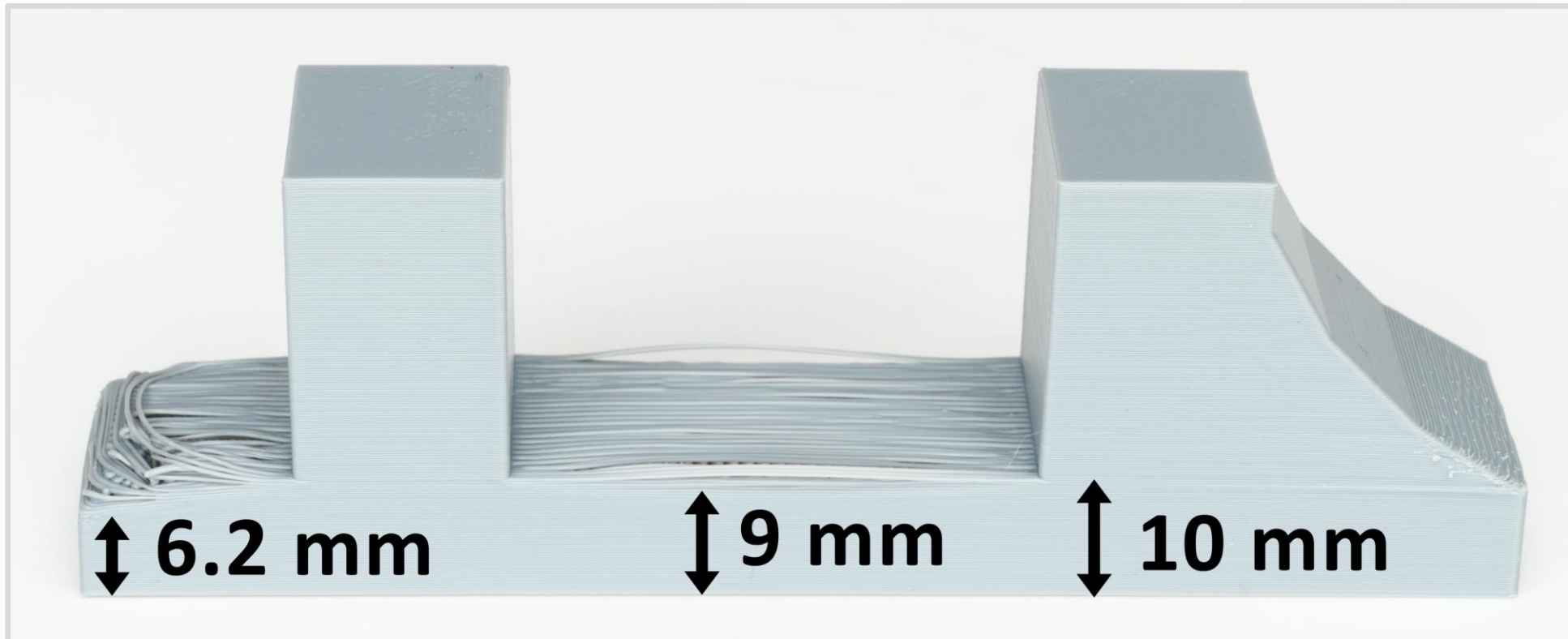
Floating Extrusion

- Extrusions that hang in the air.
- At least one end is not anchored to a solid surface.
- Advice?
- **Avoid. Very unreliable.**



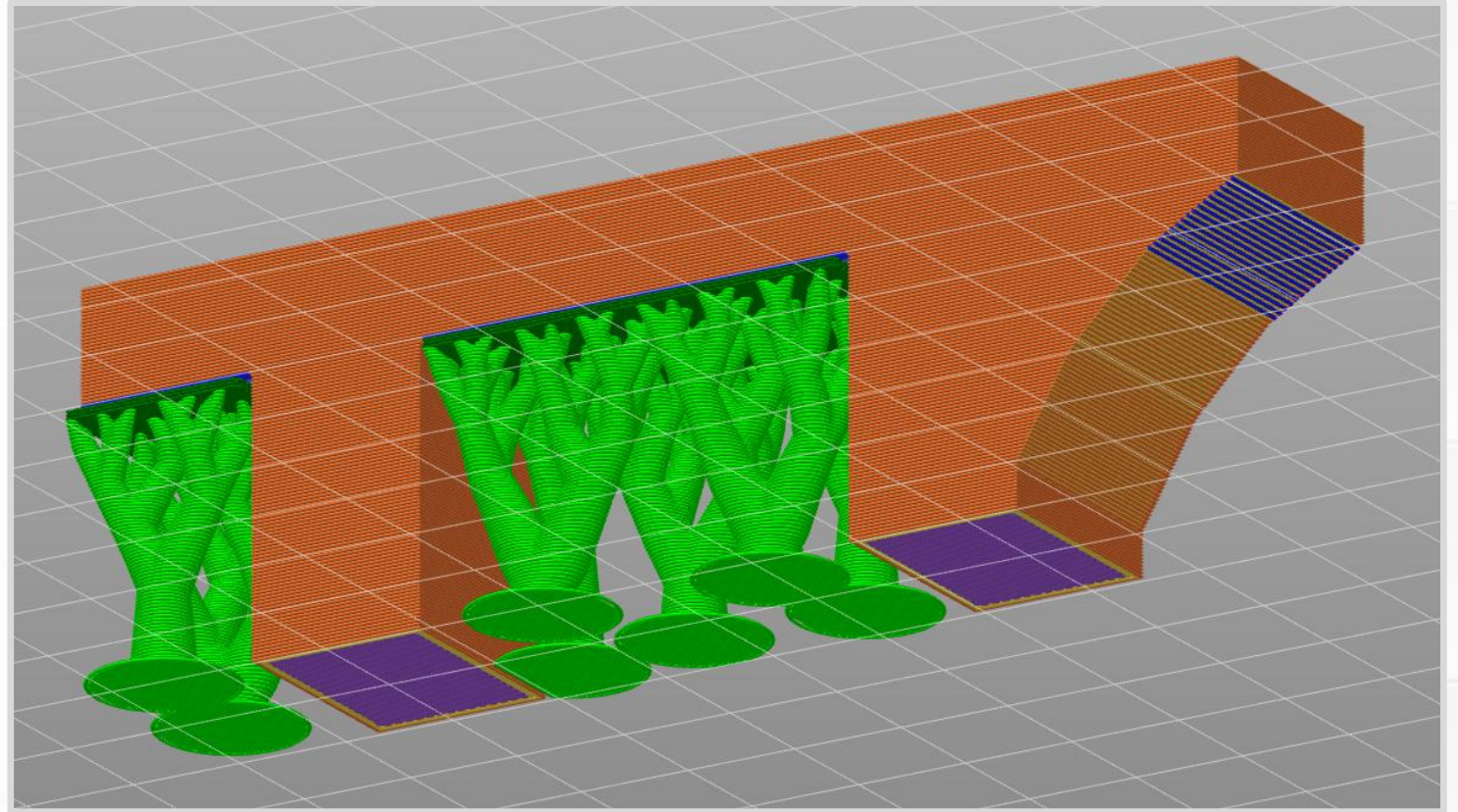
Stress Test Results

- All overhang angles look good.
- Bridge recovers to flat layers in 1 mm (**5 layers**).
- Floating extrusions recover in 3.8 mm (**19 layers**).



Supports

- Supports. Temporary scaffold that fixes:
 - Floating extrusions.
 - Long bridges.
 - Steep overhangs.
- Added by the slicer.
(Green trees-like branches.)
- Extra complexity, plastic, and print time.

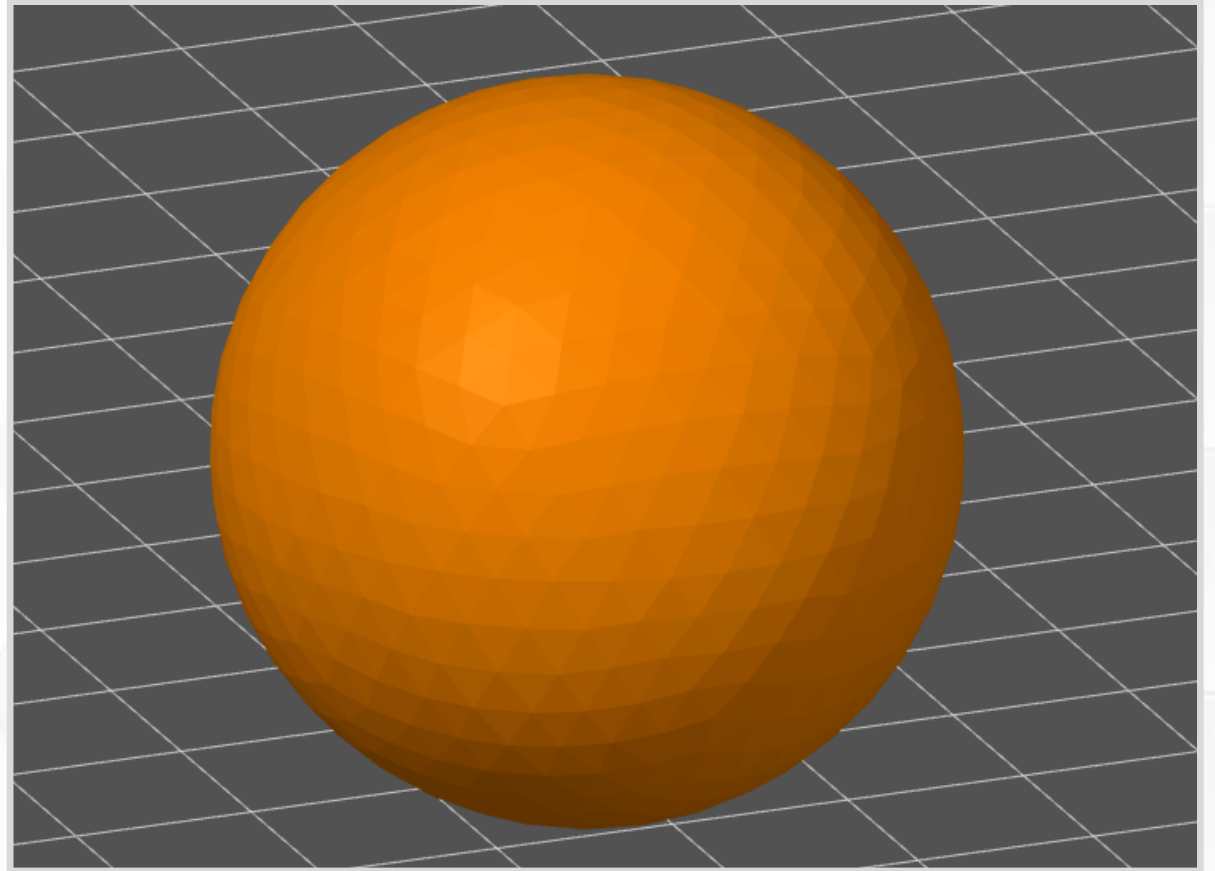


Supports

- In special situations, you can include supports as part of a 3D model.
- Tree/organic supports can reach into difficult spaces.
- Supports are a delicate balance. Why?
 - If they do not bond well enough ... they hold nothing in place.
 - If they bond too well ... they become part of the print.
- Supports can leave a rough surface finish.
- Sometimes you need to cut away pieces of support.

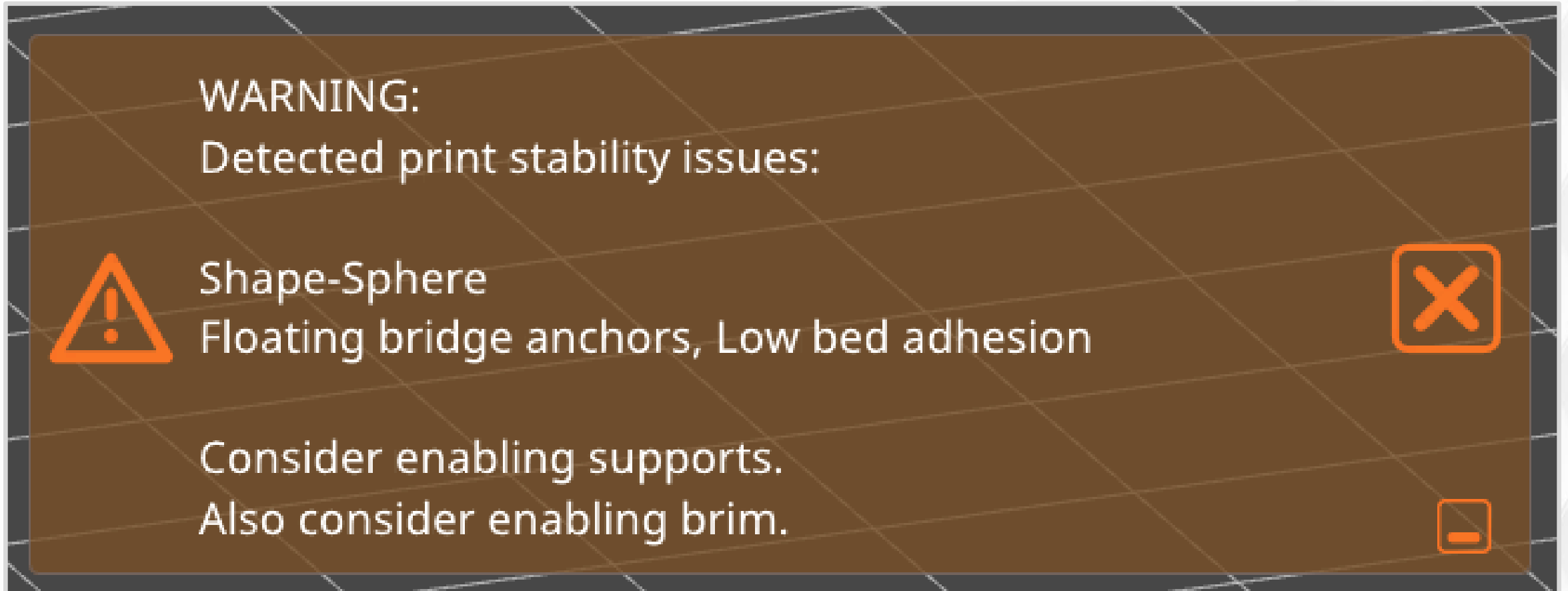
Limitations Review

- Printing a ball runs into some FFF limitations.
 - Very bad **first layer** contact. It will make spaghetti.
 - Steep overhangs on the bottom.
 - Second layer is steep enough to be a floating extrusion.
 - Stair-stepping creates uneven surface finish.



Limitations Review

- PrusaSlicer recognizes these problems.



Limitations Review

How can we fix this print?

- Cannot fix stair-stepping by changing print orientation.
 - Use small layer heights to reduce visual artifacts.
 - Use adaptive layer heights to make the surface look more uniform.
- Brim is marginally helpful.
 - Object starts with too little surface area.
- Add supports for most of the bottom.
- Print two halves, then put them together.

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Dimensional Accuracy

- Do not design parts with exact dimensions.
- This is a trap for beginners.
- Why?
 - Your prints are not exact.
 - All measurements come with **error bars**.
 - The fit is probably too tight to work as intended.
- Where do these errors come from?
- And how do we fix it?

Dimensional Accuracy

- Sources of error include (but are not limited to)
 - Shrinking and warping.
 - Inconsistent extrusions.
 - Misaligned parts.
 - Bent parts from shipping damage.
 - Incorrect belt tension.
 - Backlash, slop, or loose parts.
 - Moving parts that eventually wear down.
 - Unoptimized slicer settings.
 - Thermal expansion (discussed in Part 3).
 - Vibrations (discussed in Part 3).

Modeling Calibration

- $(Printed\ size) = (Size) \times (Scale\ factor) + (Constant\ offset)$.
- Simplified model.
 - One dimension, like X or Y.
 - Assume errors come from filament and extrusion.
- Scale factor.
 - Error changes with size.
 - Part shrinkage.
- Constant offset.
 - Error is always the same.
 - Extrusion inconsistencies. Over and under-extruding.
 - Surface roughness. Filler or additives like carbon fiber, wet filament.

Scale Factor

- Use scale factor with ...
 - Large parts (over 6 inches, 150 mm).
 - Filaments that shrink (like ABS).
- Adjust **XY scaling** or **shrinkage compensation** setting in your slicer.
- Not important for small parts.
- I often ignore XY scale factor in my prints.

Constant Offset

- Very important for any project with assembled pieces.
- **Add clearance (air gap) between parts that touch.**
- Best to add clearance while designing 3D objects.
 - XY offset settings in a slicer are less effective.
- I think about XY constant offsets in all of my designs. Very important!

Adding XY Clearance

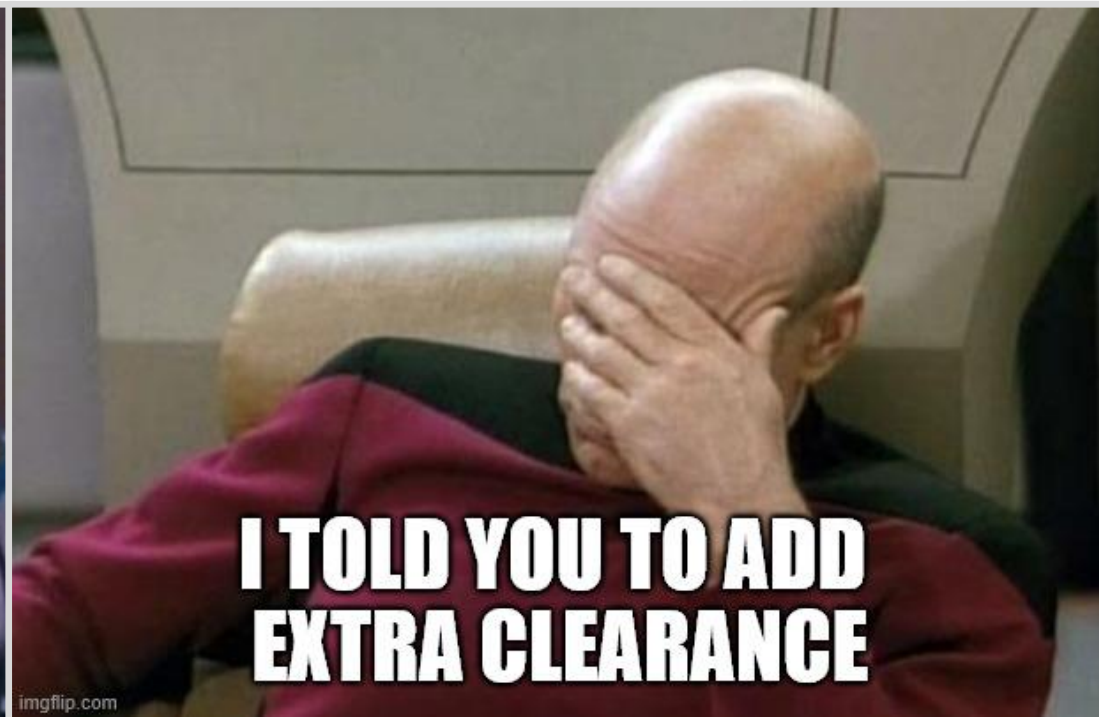
- How much XY clearance?
- It depends, but not very much for modern printers.
- *Example:* Making a hole for a bolt.
 - Add 0.2 - 0.3 mm of clearance for a close fit.
- *Example:* Making a lid that easily slides into place.
 - Add 0.4 - 0.8 mm of clearance for a loose fit.

Adding Z Clearance

- How much Z clearance?
- It depends, but less important than XY clearance.
- One layer or less.
 - 0.1 mm.
- Factors that increase clearance.
 - Bridges.
 - High and low spots on print bed.

Recommendations

- You sometimes need scale factor.
- You always need constant offset.



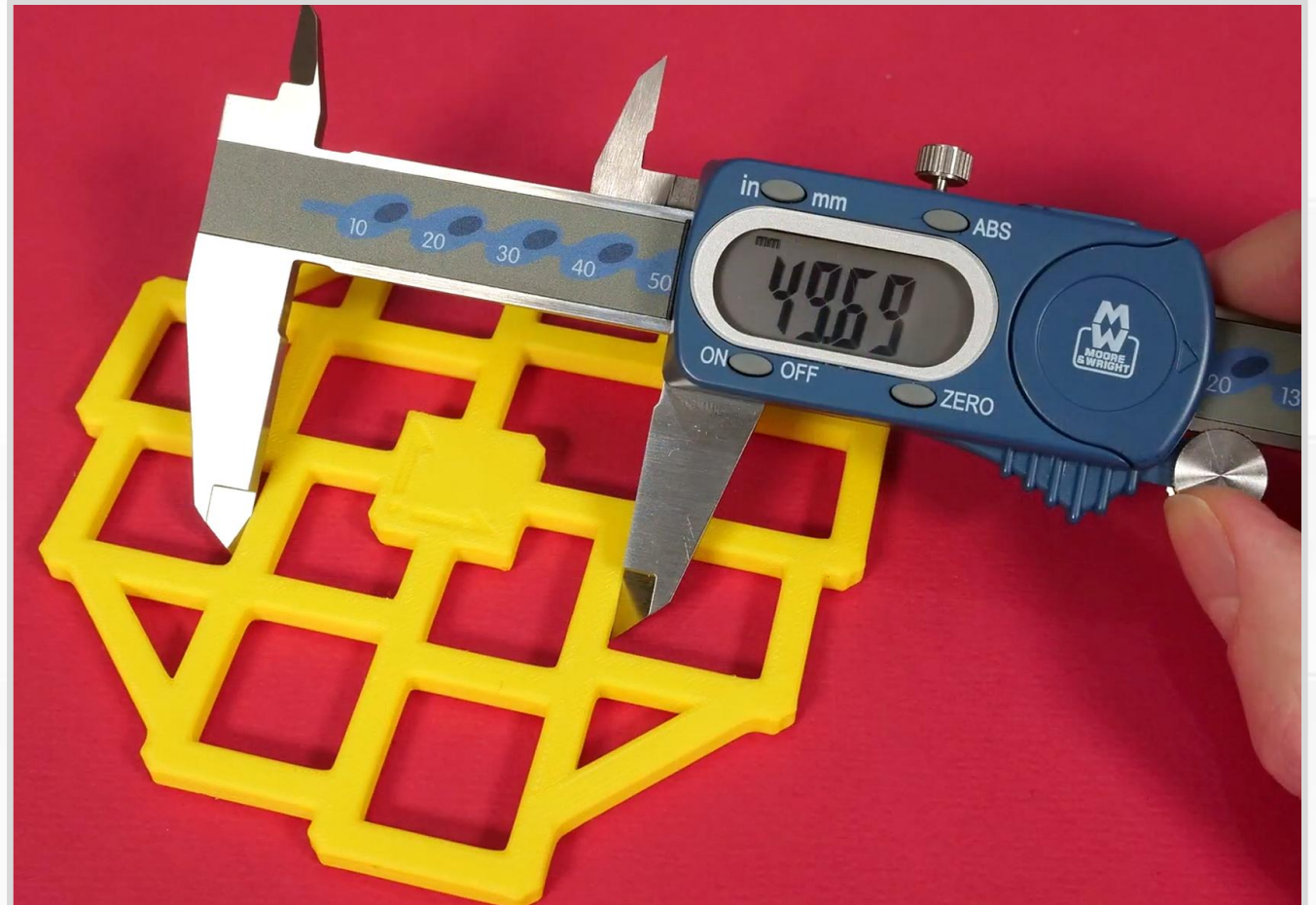
Recommendations

- *Caution:* Calibration prints cannot fix everything.
- Calibration cubes.
 - 20 mm cube. Labels for X and Y-axis.
 - Constant offset for outside dimensions.
- Clearance test by 3DMakerNoob.
 - Print-in-place clearance/tolerance.
 - Alters the gap between pieces.
 - See which pieces can move freely.



Recommendations

- Califlower by Vector3D.
 - Inner dimensions.
 - Outer dimensions.
 - Shrinkage.
 - Skew. (If your rectangles turn into parallelograms).



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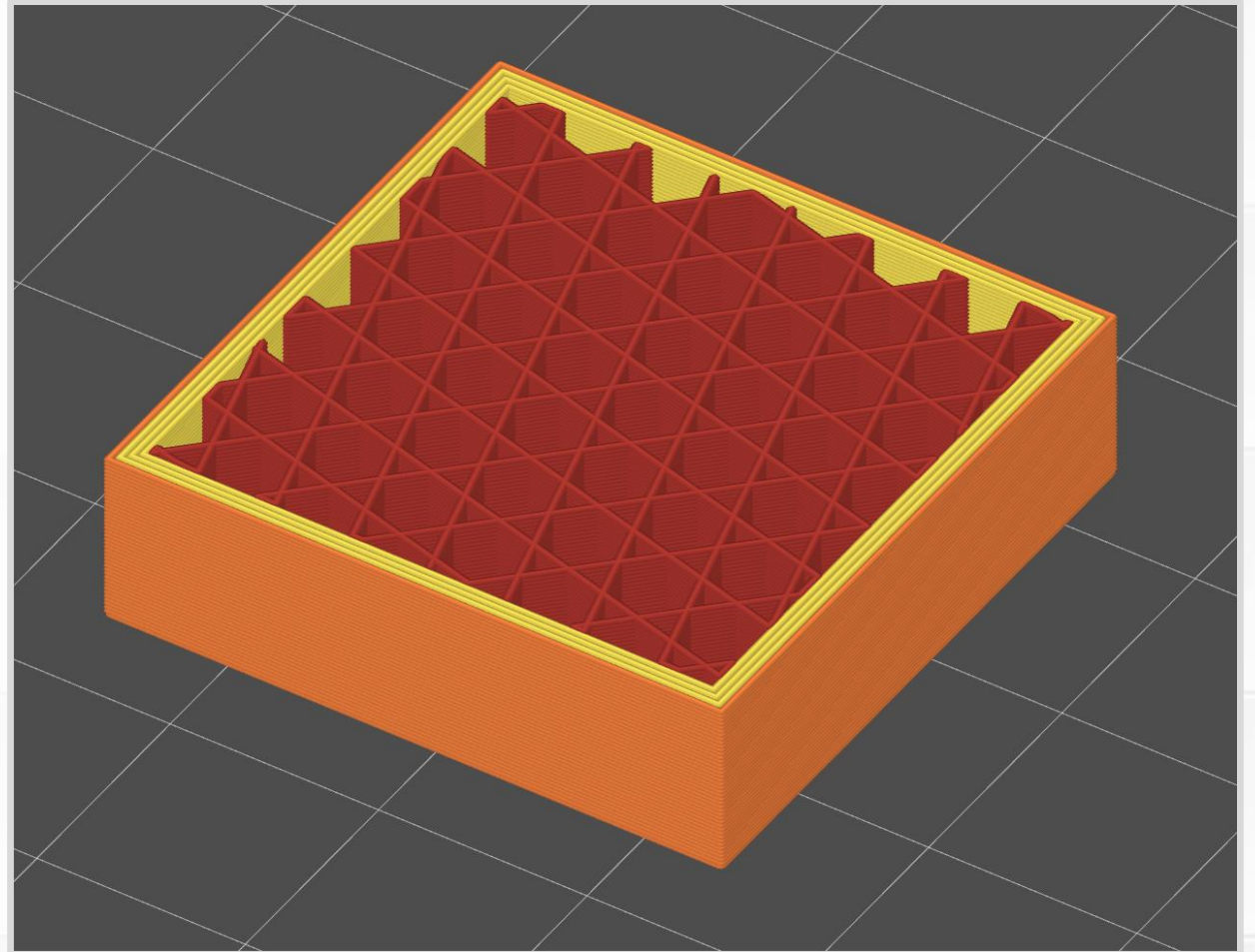
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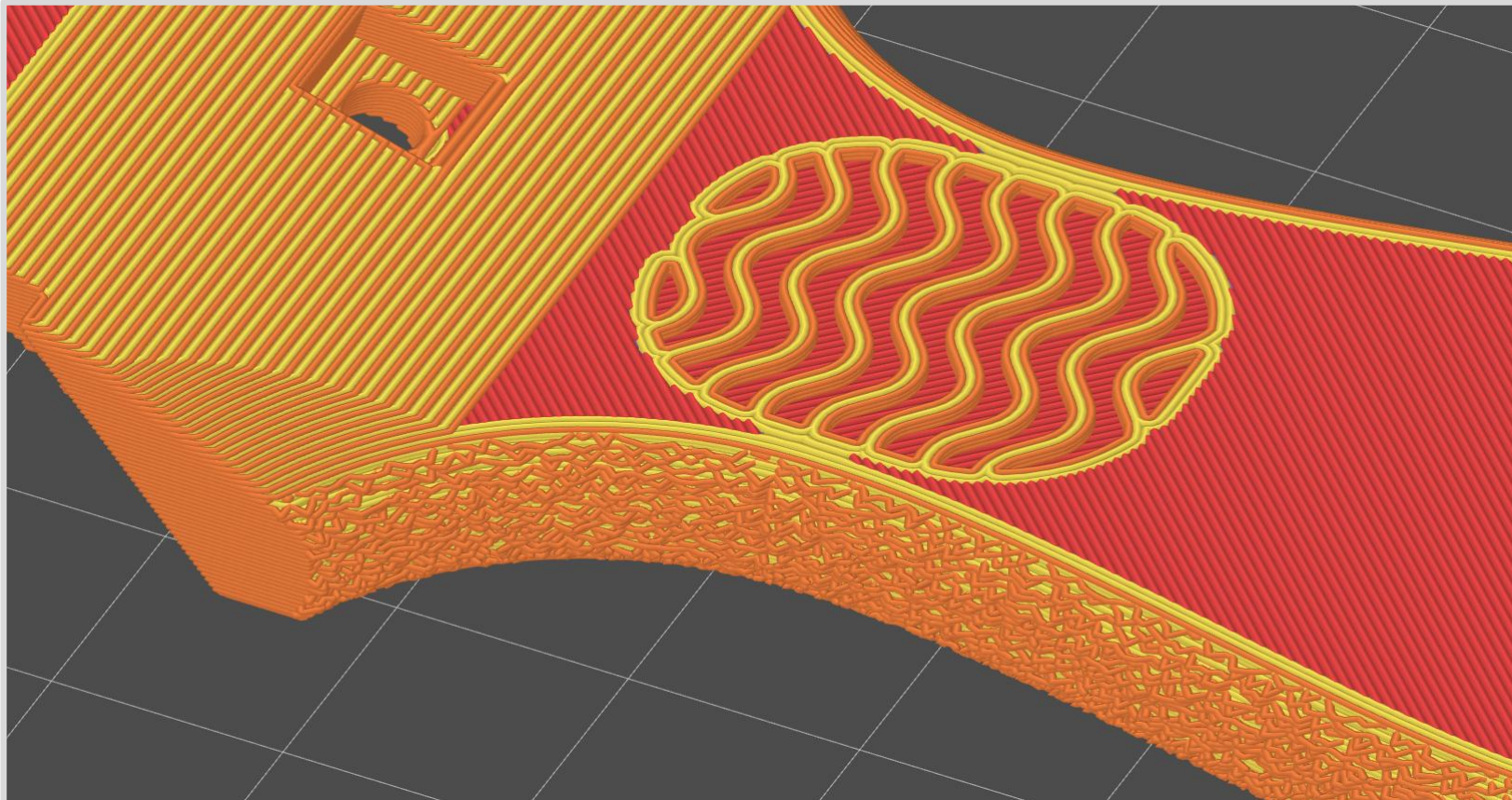
FFF Advantages

- Printers use tiny “building blocks.”
- Typical extrusions.
 - **0.2 mm tall** (1-2 pieces of paper).
 - **0.4 mm wide**.
- Not limited to uniform prints.
 - *Example:* Solid exterior and sparse infill.
- Great at making hollow objects.
 - Imagine making this shape with wood or metal – much harder.



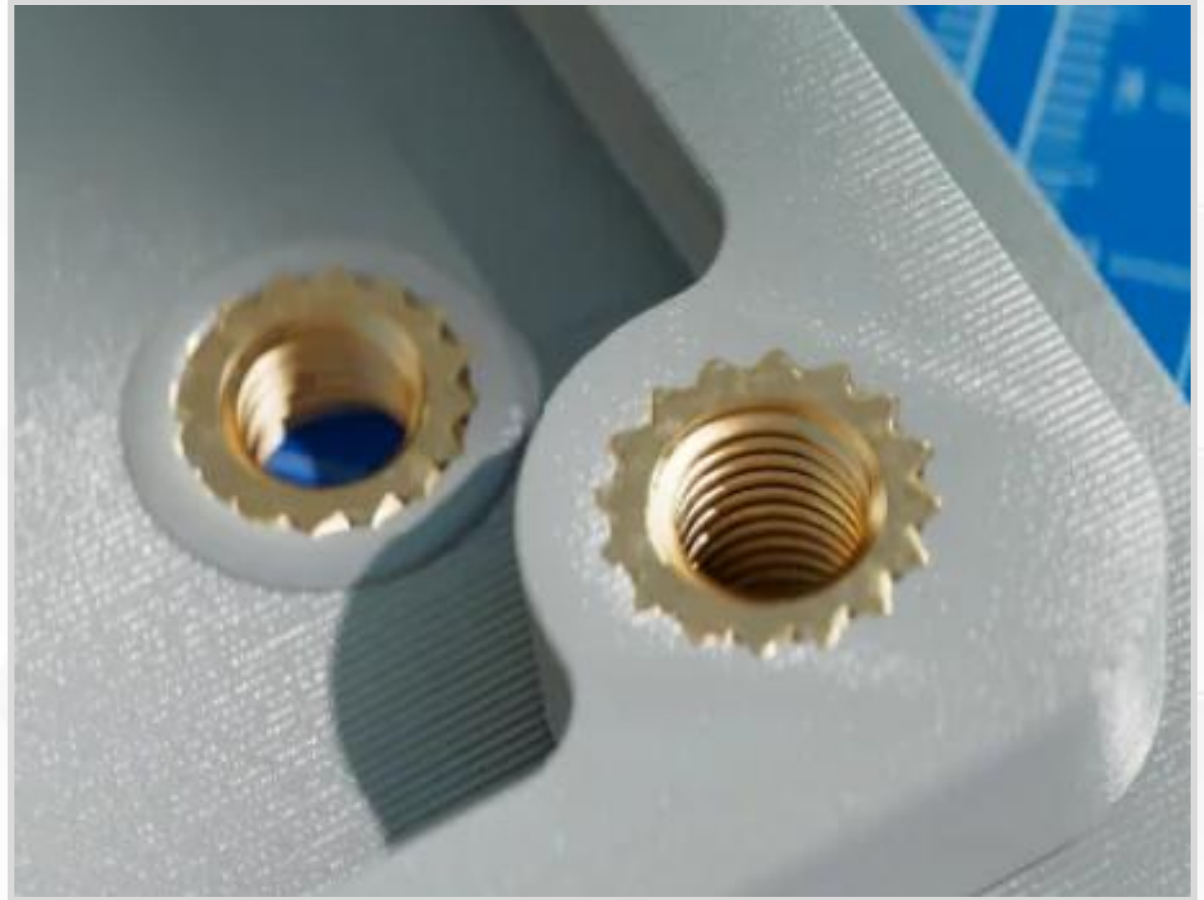
Custom Slicing

- Change slicer settings for specific regions or surfaces.
- *Example:* Add texture to handles with fuzzy skin (bumpy extrusions).



Heat Set Inserts

- Plastic is soft → Screws easily wear out plastic holes.
- Heat set insert. Threaded anchor for screws and bolts.
- Install with a hot soldering iron.
- Locks in place when it cools.
- Very durable.



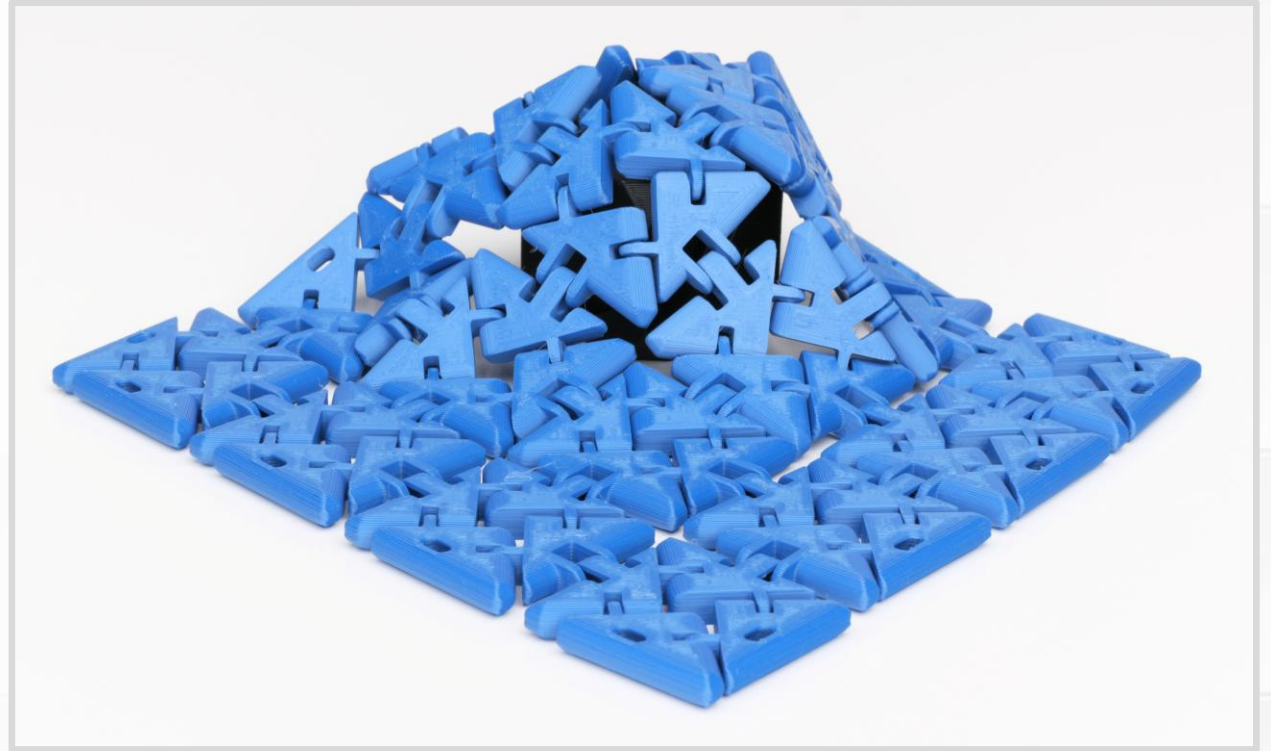
(Pictures from [CNC Kitchen](#))

Captive Parts

- Insert an object into a pocket or void in the middle of a print.
 - Nuts and fasteners.
 - Magnets.
 - Weights.
 - Reinforcing plate
- Object is locked inside when the print finished.
- Add a pause command in the Gcode.
- Object must sit below the current layer, or it creates a collision with the toolhead.
- Waiting around for the right layer is inconvenient.

Print-in-Place (PIP)

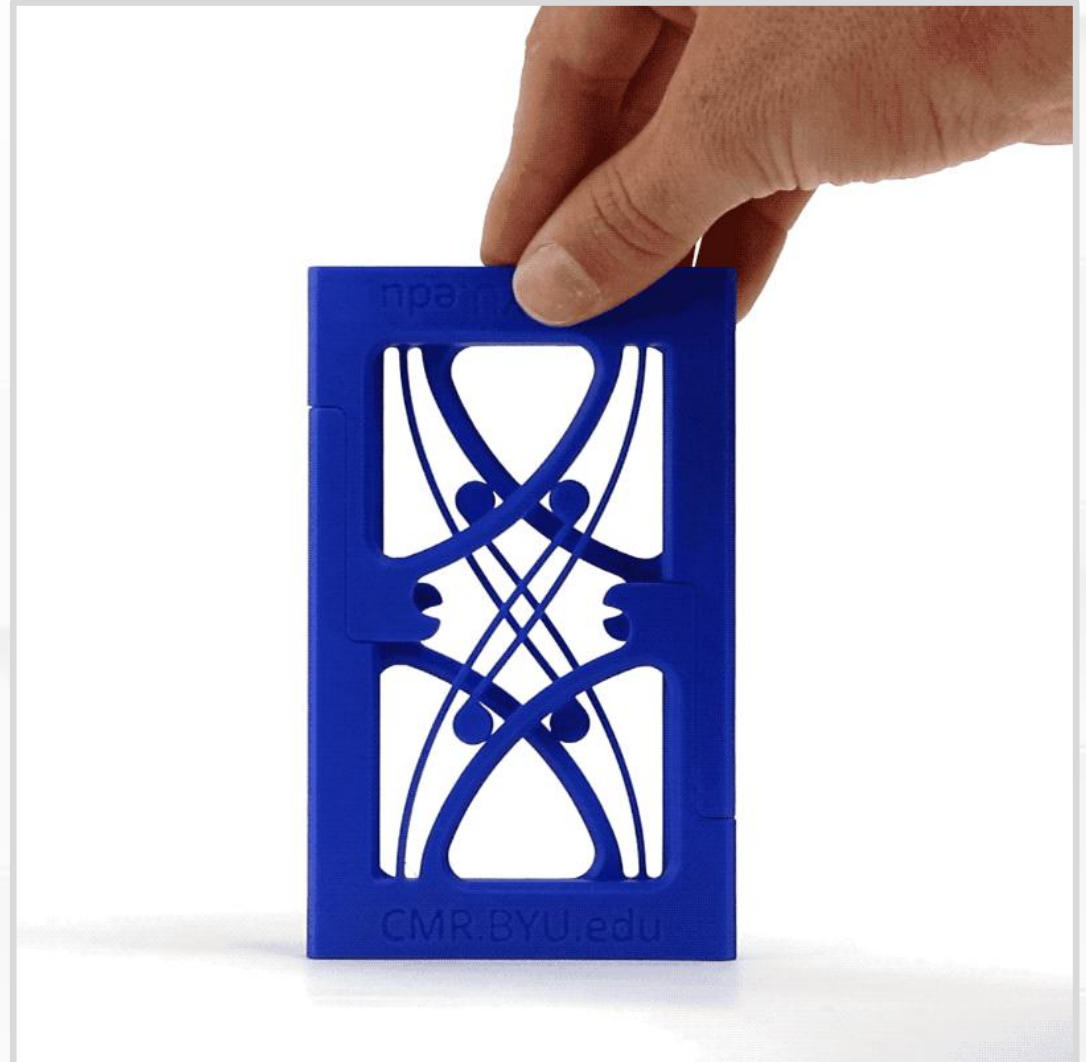
- Single print with interlocking parts.
- Assembled when the print finishes.
- *Example:* Hinges, chain links, chainmail.
- Flexify collection by [printables.com/@Flying Gyroscope](https://printables.com/@FlyingGyroscope)



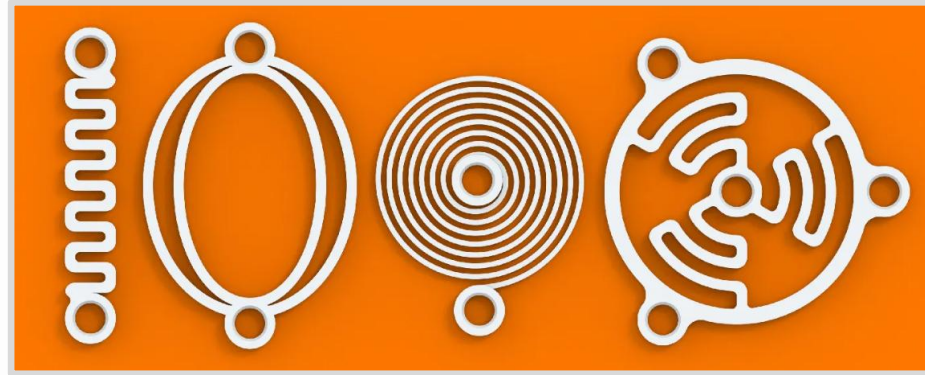
Functional Flexing

- Springs and **compliant mechanisms**.
- Features/shapes that function by bending and flexing.
- Elastic materials (springy, without permanent deformation).
- Squeezing springs is literally stressful.
 - Remember strength and layer lines.
 - Flex springs in XY plane for maximum durability.

(Animation, Constant Force Mechanism by BYU CMR)



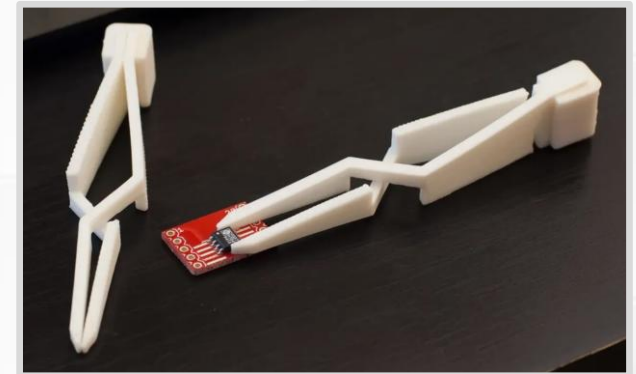
Springs by VC Design



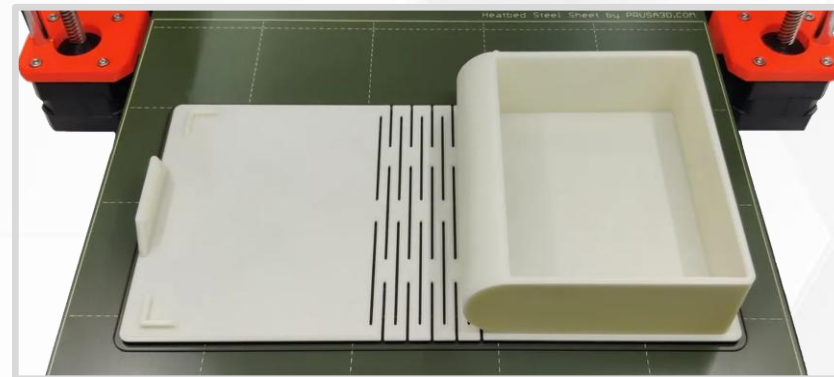
Clip by AE



Cross tweezer by joo



Living hinge by DrJones



THANK YOU